Client internals and version profiles, part 4 of 4

Client internals and version profiles, part 4 of 4

This part contains 29 notes, from `client-internals/saved-passwords.md` through `version-profiles/version-coverage-2026-10-24.md`.

`client-internals/saved-passwords.md`

Saved passwords and the client's DES string cipher

Date: 2026-10-21

Traced from client 2.05 rev 2. Tags: **[2.05]** read from the client · **[test]** checked with `tools/file-formats/graaldes.py`.

Warning for anyone holding old installs

The client stores account passwords in the registry under `HKEY_CURRENT_USER\Software\Graal\Client` (`password`, and `<account>:pass` per account). They are DES-encrypted with an 8-byte key that is the same in every copy of the client: `~2%$zsd^`. **[2.05]**

The same key and string form are already used by 1.3 test (Nov 1999): a value saved by 1.31/1.32 in a lab run decrypts with `graaldes.py`, and the key string is in every client from 1.3 test to 1.39 (`live-trace-early.md`). So the warning covers those old installs too.

That means anyone who can read the registry, or a `.reg` export or backup of that key, can recover the password with no secret at all. Treat these values as plain text. If an old machine, disk image or exported profile is shared, delete the values first, or tick "Don't save the passwords" in More Options (the client then deletes them on close).

Cipher

- DES in ECB mode, standard tables, with one twist: **each key byte is bit-reversed** before the key schedule (bit 0 becomes bit 7). This is how the Acme Labs Java `DesCipher` handles keys, and the client's DES unit behaves the same. **[2.05] [test]**
- Equivalent: standard DES with the key `bitreverse(k[0]) … bitreverse(k[7])`.

String format (48 characters)

Encrypt: 1. Truncate the plaintext to 23 bytes. 2. Build 24 bytes: `len`, the plaintext, then random padding bytes. 3. Encrypt as three DES blocks. 4. Write each of the 24 bytes as two characters: `(b >> 4) + 0x20`, `(b & 0x0F) + 0x20`.

Decrypt reverses this. Input shorter than 48 characters gives an empty string. The first decrypted byte is the length, and values of 0 or ≥ 24 give an empty string. **[2.05]**

Because the padding is random, the same password encrypts to a different string each time.

Other uses of the same cipher

- The hidden listserver host: the server list frame builds a different 8-byte key and a 48-character ciphertext one character at a time. It decrypts to `<list-host-classic>`, port `14921`. **[2.05] [test]**
- The web-level cache uses the same DES routine with per-block keys. See `formats/gwebl001-cache.md`.

Tool

`tools/file-formats/graaldes.py` (pure Python):

```
python3 graaldes.py '<48-char value>'
```

It implements `GraalDES` (block cipher), `string_decrypt` and `string_encrypt`. It reproduces the listserver string and round-trips its own output.

`client-internals/sign-panel.md`

The sign panel and NPC drawing (2.05)

How the 2.05 client draws a sign while the player reads it, the keys that turn its pages, and what it draws for an NPC whose image is missing. Read from the 2.05 program and checked against the `letters.png` sheets of 1.39 and 2.22. Seihai follows this in `documentation/signs.md`.

The sheet

`letters.png` is 256×224: 16 glyphs a row, each 16 wide and 32 tall, so glyph `g` is at (`g % 16 × 16`, `g / 16 × 32`). It carries the font, the button pictures and the pieces of the scroll frame. The client loads it once, at start.

Glyph order: a character's glyph is its place in

```
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789!?-.,#>()#####"####':/~&### <####;
```

(then a newline). The # places are glyphs with no plain character. Each glyph has its own advance, the client's table of half widths doubled (I is 6 pixels, space 8, most letters 12).

glyphs	use
96, 97, 98, 99	frame corners: top-left, top-right, bottom-left, bottom-right
100, 101	top and bottom edge
102, 103	left and right edge (32 pixels tall)
91 to 94	the A, B, X and Y buttons
77 to 80	the up, down, left and right pictures
71 to 75, 87, 88, 67	the other button pictures

The pixels at (252..255, 223) are colours, not art. The client fills the inside of the frame with (254, 223) and, 8 pixels further in, (255, 223), and skips a fill whose colour is the sheet's transparent one. On the 2.22 sheet both are the same cream; on 1.39 the inner one is white.

The panel

- 24 cells wide and 9 tall (16-pixel cells), centred across the view.

- On a 640-wide view it sits at y 256, or at y 128 when the player is more than 14 tiles down the view. At 800: 18 tiles, y 320 or 160. At 1024: 22 tiles, y 448 or 192. Which two values the client compares is inferred from the code's shape (the player's row against the view's top); the thresholds and positions are as read.
- The frame: the top and bottom edges every 16 pixels, the sides every 32 pixels, the corners over them, then the two fills from (x + 16, y + 24) to the right and bottom pieces.
- Text: three lines a page, from (x + 16, y + 16), 32 pixels apart. The sign's text is split at newlines.
- In a line, # and a letter draws button pictures: `#A`, `#B`, `#X`, `#Y`, `#u`, `#d`, `#l`, `#r`, `#h`, `#x`, `#y`, `#z`, `#4`, `#..` # and any other letter draws glyph 86 and moves 16 pixels. `#i` and a character puts up an image from the level's list.
- A character outside the glyph order is drawn by the client in its system font, in the colour at (253, 223).

Keys while reading

Reading a sign is its own input state: walking and the A / S / D handler are only reached outside it, so the player does not walk, swing, lift or use an item while a sign is open. Switch-to-chat still works, and while the chat bar has focus the sign keys are skipped. **[2.05]**

- The map key ends reading at once.
- The first direction key (Up) turns back three lines, when not on the first page.
- Every other game key (Left, Down, Right, D, S, A, item list, pause) turns to the next three lines, unless the page on show is not full (the sign has fewer lines than the page's first line plus three); then reading ends. A sign of exactly three lines therefore shows an empty second page before it closes, as in the client.
- A key counts only when no game key was down on the tick before, so a held key does not repeat and the key that walked into the sign does nothing until it is let go.
- The client plays a sound when the page turns and runs `playerendsreading` when reading ends.

(An earlier version of this note said any other key ends reading; only the map key does.)

NPCs with no image

- In game, an NPC with no image loaded draws nothing of its own, only its text if it has any.
- An NPC with a GANI draws the GANI, and a character NPC is drawn the way players are.
- The client's placeholder picture for an NPC with no image appears only in the level editor's view.

Confirmed, inferred, not tested

- Confirmed from the program: the glyph grid, glyph order and advances, the frame pieces and fill colours, the panel size and line layout, the # codes, the paging rule, and the NPC draw rules above.
- Inferred: which two values the panel's height test compares, and that key 0 is Up.
- Not tested against a real client screenshot.

`client-internals/sound-and-music.md`

Sound and music: 2.05 rev 2, compared with 2.17 / 2.22

Date: 2026-10-21

The 2.05 part is derived from the 2.05 rev 2 binary (sound path, options reader, GS1 sound commands). The later-client part comes from imports, option keys, and strings of `Graal_2171.exe`, `Graal_222.exe`, and `Graal_231.exe`. Names only, no code. Tags: **[2.05]** · **[2.17]** · **[2.22]** · **[inf]**.

2.05 at a glance

What	Backend
Sound effects (`.wav`)	DelphiX `TDXSound` / `TDXWaveList` on DirectSound
Music, Windows MCI path	invisible VCL `TMediaPlayer` (MCI, `mciSendCommand`)
Music, Winamp path	`TWinamp` component that remote-controls a running Winamp 1.x/2.x
Music volume	`midiOutSetVolume` (MCI path) or Winamp IPC messages

Options (`graalconfig.txt`)

Key	Meaning
`musicon`	play music at all
`usewinamp`	send music to Winamp instead of MCI
`soundfxon`	play `.wav` effects
`reversestereo`	flip the pan sign

The backend is chosen when the game starts and again whenever the More Options dialog closes. Switching stops the current track, frees the old backend, and creates the new one. Turning music off stops the current track. **[2.05]**

One entry point for every sound

All sound requests go through one routine that takes `(file, level-offset x, level-offset y, x, y)`. The GS1 commands, GANI `PLAYSOUND`, item pickups, bombs, swords, and baddies all call it. About two dozen call sites use it. **[2.05]**

1. **Loop prefix.** A leading `+` means "looped". The `+` is stripped. Looped effects are added to the player's looped-sound list so that they can be stopped later.
2. **Missing file.** If the file is not on disk and the client is online, it sends a file request to the server (the same path that fetches images). The sound plays on a later trigger once the file has arrived. **[inf]**
3. **Position.** The listener is the player at (x + 1.5, y + 2). A sound from a neighbouring gmap level is shifted by ±64 tiles first.
    - Pan = (sound x − listener x) × 150, clamped to ±10000 (DirectSound pan units). `reversestereo` negates it.
    - Volume = a negative attenuation that grows with the distance on both axes, clamped at −10000 (DirectSound hundredths of a dB).
4. **Dispatch by extension.**
    - **`.wav`** → effect path (below), only when `soundfxon`.
    - **anything else** → music path (below), only when `musicon`. `.mid` is the common case.

Effect path (`.wav`)

- Fifteen effects are known by name and flagged on use: `arrow`, `axe`, `bomb`, `crush`, `extra`, `fire`, `horse2`, `item`, `lift`, `lift2`, `put`, `steps`, `sword`, `thunder`, `water` (`.wav`).
- At most **8** effects play at once. A ninth request is dropped.
- Each file is loaded once into the wave list and cached by name. Later plays reuse the buffer with new pan, volume, and loop settings.

- `stopsound file` finds the cached wave by file name (the path is ignored) and stops it.
- Leaving a level or entering the editor stops every looped effect in the player's list.

Music path: MCI

- An invisible `TMediaPlayer` is created on the main form with automatic device type, so `.mid` goes to the MCI sequencer. Other types depend on the MCI drivers installed on the machine. **[inf]**
- If the requested file differs from the current one, or nothing is playing, the client sets the file name, opens it, and plays.
- For `.mid` the player's completion notification is turned on. A handler on the main form replays the track when it ends, so **MIDI music loops forever** until `stopmidi` or another track.
- `setmusicvolume left,right` (0..1 each) calls `midiOutSetVolume` with left and right scaled to 0..65535. That is the system MIDI-out volume, not a per-track volume.

Music path: Winamp

`TWinamp` is a third-party Delphi component (unit `Winamp`). It controls an external Winamp through the public Winamp window-message API. **[2.05]**

- It finds the player by window class `Winamp v1.x`. If no Winamp is running, it reads the Winamp path from the registry key `Software\CLASSES\Winamp.File\shell\Play\command` and starts it.
- A timer polls the window and status, and fires events on play, stop, pause, title change, and exit.
- Playing a track: if Winamp's current file differs or Winamp is stopped, the client loads the file and sends *play* (IPC_STARTPLAY / the play command).
- `stopmidi` sends *stop* when Winamp is playing or paused.
- `setmusicvolume left,right` maps to Winamp volume `(left + right) / 2 × 255` (IPC_SETVOLUME). The balance maps to Winamp panning.
- The component can also list the formats Winamp's input plugins accept (`winampGetInModule2`). In Winamp mode, music is effectively "whatever Winamp can play", MP3 included.

GS1 sound commands in 2.05

Command	Effect
`play file`	One-shot at the NPC's position (NPC x/y + half its image size).
`playlooped file`	Same, with the + loop prefix.
`stopsound file`	Stop a cached effect by file name.
`stopmidi`	Stop the current music (MCI or Winamp).
`setmusicvolume l,r`	MIDI-out volume (MCI) or Winamp volume and pan.

2.17 / 2.22 / 2.31: BASS replaces DirectSound and Winamp

All three later clients import **`bass.dll`** with the same set of functions: init/start/free/update, sample load/play/stop/free, tracker music load/play/free, file streams, **URL streams**, stream length/position, channel attributes, and the **recording** API. `winmm` is still imported, now with `midiOutGetNumDevs` / `midiOutGetDevCapsA` added. **[2.17][2.22][2.31]**

Area	2.05	2.22
Effects	DelphiX DirectSound wave list, 8-voice cap	BASS samples **[inf]**
Music files	MCI (MIDI) or Winamp (anything Winamp plays)	BASS streams and tracker modules. MIDI still via MCI **[inf]**

Area	2.05	2.22
Internet radio	none	BASS URL streams, `radio` / `radiovolume` **[inf from names]**
Winamp	`usewinamp`, `TWinamp`	**removed**: no Winamp component, key, or window class
Volume options	none (only `setmusicvolume`)	`midivolume`, `mp3volume`, `radiovolume`, `sfxvolume`, `voicevolume`
Script access	`play`, `playlooped`, `stopsound`, `stopmidi`, `setmusicvolume`	same commands plus `musicpos` / `musiclen` (stream position and length)
Voice chat	none	microphone capture through BASS recording, `hvdi.dll` codec

Voice chat (2.22)

New options: `microon`, `microinputdevice`, `microactivatebyvolume`, `microactivationlevel`, `microactivatekey`, `voicecodec`, `voicevolume`. Status strings: "Could not initialize Microphone", "Microphone enabled", "Green when microphone activated". An `emoticonmicro.png` icon is shown over a talking player.

`hvdi.dll` is the HawkVoice voice library (exports `hvdiPacketEncode` / `Decode`, `VOX`, `AGC`, codec selection). The client captures with BASS, encodes with HawkVoice, and sends the voice packets over the network. **[2.22][inf for the transport]**

`basswma.dll`

The 2.2.2 folder ships `basswma.dll` (the BASS WMA add-on), but its use depends on the version:

- **2.17.1** loads it at run time by name and uses `BASS_WMA_StreamCreateFile` / `StreamPlay` / `StreamFree`. **[2.17]**
- **2.22**, **2.31**, and GraalEditor 2.3.1.1 neither import nor name it. In those builds the DLL is a leftover and WMA music is not supported. **[2.22]**

Which action plays which sound

See `live-checks-editor-offline-sound.md` for a table of engine sounds by action, partly confirmed in a running 2.05 client.

Open

- 2.22 extension-to-backend dispatch (which types go to samples, modules, or streams) needs the 2.22 sound routine traced.
- Exactly what 2.05 does after a missing sound file arrives from the server.

`client-internals/status-bit-0x20.md`

STATUS bit 0x20: the sword is hidden (2.05)

The player STATUS property (prop 18) carries bit 0x20 while the `hidesword` timer is running. Read from the 2.05 program.

How the bit is set

- The script command `hidesword n` takes one number. It is clamped to 0..30, multiplied by 20, and stored as a tick count (the game tick is 1/20 s, so `n` is in seconds).
- Each game tick, while the count is 0 or more, it goes down by one. It reaches -1 one tick after it reaches 0.
- When STATUS is built for sending, bit 0x20 is set when the count is 0 or more.
- The count starts at -1 when the player object is made, and is set back to -1 when the player is reset.

`hideplayer n` works the same way, with the same clamp and scale, but on its own counter, which drives bit 0x02 (hidden). In the command table, `hidesword` and `hideplayer` are entries 59 and 69. Both take one number argument (`R`).

What the bit does

- Another player whose STATUS has 0x20 set (or 0x02) cannot hurt with a sword swing. The hit gate checks the attacker's STATUS (see `decomp-findings-2026-10-24-d/03-hit-gates-and-open-points.md`).
- In the local client, the counter is read only to build STATUS and to count down. Nothing else reads it, so `hidesword` does not stop the local swing itself. The swing is only refused a hit at the victim's end.

Confirmed, inferred, not tested

- Confirmed: the command, the clamp and scale, the countdown, and the STATUS bit.
- Inferred: that a client also leaves out the sword picture of a player with bit 0x20. The drawing code was not traced.
- Not tested on a wire or in the lab.

`client-internals/status-display.md`

Status display (HUD)

Date: 2026-10-28

- **Code:** read from the 2.05 rev 2 client's status routine, with two constants checked in its executable: the heart-row table and the small-heart stagger. **[2.05]**
- **Sheets and picture:** the shipped sheets, and a 2.22 lab screenshot. **[cap]**
- **What these notes leave out:** addresses, routine names and decompiler text.

Sheets

- Every build from 1.41 on ships a 224×145 `state.png`.
- The 2.22 sheet matches the 2.05 rev 2 and 2.1 test 3 sheets pixel for pixel, except for one 27×14 patch at (190, 60), which the status routine does not use.
- 1.41's sheet has three blank buttons side by side at x 158, 180 and 202.
- The 2.0-era NewMain sheets have one button at x 202, which the routine draws in all three slots.

Layout

All positions are measured from the top-left of the game view. The routine uses a base point of (15, 15).

Part	`showstats` bit	Sheet rect (x, y, w, h)	Screen position
Rupee, bomb and arrow icons	0x02	(80, 33, 120, 16), one strip	(274, 30)
Rupee count	0x04	digit d = (14d, 81, 14, 16)	1–7 digits, no leading zeros, 12 px apart, centred on x 280, y 48
Bomb count	0x08	same digits	two digits at x 326 and 338, y 48
Arrow count	0x10	same digits	two digits at x 374 and 386, y 48
Hearts	0x20	big (40·half, 0, 40, 44); small (80 + 31·half, 0, 31, 33)	see below
Buttons	0x01	(202, 0, 22, 33)	x 15, 80 and 145, y 30
Key names	0x01	text	on each button: 26 pt Helvetica, white, with a black copy at +1, +1; smaller sizes for two-letter and longer key names. Key 9 (item list) is drawn small at (147, 27)
Glove icon	0x01	sprite 145 for power 2; sprite 325 drawn on top for power 3	(40, 29); power printed at (60, 31) when it is 2 or more
Sword icon	0x01	sprite 17, taken from the player's sword image	(105, 31); sword power printed at (125, 31)
Selected weapon	0x01	the weapon's whole image	x 170, vertically centred on y 45
Alignment bar	0x40	frame (0, 97, 130, 22); fill (0, 119, AP, 10)	(15, 65); fill at +22, +6
Magic bar	0x80	frame (0, 49, 130, 22); fill (0, 71, MP, 10)	(15, 90); drawn only with weapons on and MP > 0

Bits 0x100 and 0x200 belong to the player icons and to the stacked weapon image list, which are drawn by the same routine but are not part of the status display.

- **Sprite rects.** Sprites 17, 145 and 325 come from the sprite table in `arrays.dat` (array 24). The 2.22 file gives:
  - 17: (94, 36, 28, 28), from the sword image
  - 145: (28, 212, 28, 32), from `sprites.png`
  - 325: (0, 212, 28, 32), from `sprites.png`
- **Weapons flag.** The same flag gates the S and D keys. With it off, only the first button, its key name and the alignment bar are drawn.

Hearts

- **Drawing.** One heart is drawn per container while power remains; empty containers are not drawn. Power is a value in hearts: heart *i* is full when `power − 1 ≥ i` and half when `i < power`.
- **Big hearts.** Used while power is 4 hearts or less. They start at (W − 198, 24), 45 px apart, where W is the view width.
- **Small hearts.** Used above 4 hearts, 35 px apart. New rows start at hearts 5, 11, 16, 22 and 27. Row *k + 1* is 20(k + 1) px lower and shifted `round(17.5(k + 1)) − 35` px sideways. The whole block moves 10 px down while power is below 6 hearts.

Not read

- The 2.22 and 2.31 status routines. Their sheets and the 2.22 picture match 2.05.
- The routine of any build before 2.05, including how 1.41 uses its three button cells.
- The text drawing that prints the glove and sword power: its font, colour and anchor.

The selected weapon (2.05 rev 2)

The image beside the D button belongs to the selected entry of the player's weapon list.

- **One list.** Server weapons (`PLO_NPCWEAPONADD`) and built-in weapons (`PLO_DEFAULTWEAPON`) share one list, in arrival order. A built-in weapon is skipped when the list already has its name.
- **Adding.** Every add selects the new last entry.
- **Selecting index *i*.** The client takes the first entry from *i* on, wrapping round, whose name is not empty and does not start with `-`. If every entry is hidden, *i* stays selected.
- **Deleting.** The selection keeps its index if that index still exists. Otherwise it moves to the new last entry.

Built-in weapons

The client keeps a 25-entry item table. Only the entries with a weapon script become weapons. The weapon's name is the item name with its first letter in upper case.

id	item	image (2.05)
7	bow	`wbowi1.gif`
8	bomb	`wbomb1.gif`
12	superbomb	`wbomb2.gif`
20	fireball	`wbowi2.gif`
21	fireblast	`wbowi3.gif`
22	nukeshot	`wbowi4.gif`
23	joltbomb	`wbomb3.gif`

- **The other ids:** greenrupee, bluerupee, redrupee, bombs, darts, heart, glove1, shield, sword, fullheart, battleaxe, goldensword, mirrorshield, glove2, lizardshield, lizardsword, goldrupee and spinattack. They are pickups, not weapons.
- **2.22:** in the lab the client asked for the same names as `.png`.
- **Id 8 and `wbomb1`:** the 2.05 table ties them together. The lab capture could not tie them, because 2.22 asked for `wbomb1.png` whether or not id 8 was sent.

`client-internals/v2.00-connect-path.md`

2.00: list reply, Connect and the grey window

Date: 2026-10-30. Build: Graal v2.00 (`GNW24011`, md5 `c93e659e2cda36379aaea4706db20b38`). Static reading of the exe, checked against the cross-laptop lab of 2026-10-31 (seihai `documentation/cross-laptop-2026-10-31.md`). Derived facts and addresses only. Tags as in `protocol/listserver.md`: **[bin]** · **[cap]**.

The answer

The lab 2.00 run that never reached the gserver was not a list-format problem. The client folder (a copy of the 2.22 distro) had no `objects.dat`. 2.00 checks for it when Start is pressed, sets a fatal flag, and the Connect path later returns early on that flag. The result is a blank grey game window, no socket and no DNS lookup. **[bin]** The lab saw exactly that. **[cap]** The lab listserver and GServer-v2 need no change for 2.00.

Classes

Delphi RTTI, VMT addresses: `TGraalClient` 0x4dabc8 (parent TObject), `TServerListClient` 0x4dac4c, `TAdventureClient` 0x4dacbc. `TServerListClient` overrides virtual slot 2 (0x4db984, the per-message handler) and slot 3 (0x4dbc3c, an empty `ret`). Slot 0 (0x4db160) closes the socket. **[bin]**

List reply [bin]

- 0x4db984 acts on ids 0 and 1 only. Any other id, including the 2 (status), 3 and 5 the lab listserver sends, is ignored.
- Id 0: `GCHAR count`, then per server `GCHAR field count` and that many `GCHAR len + text` fields, each server kept as a string list. The list goes to the list form (0x4a4ffc), and the handler then closes the list socket through slot 0. A row is drawn only with at least 6 fields (0x4a512c), and the detail view needs at least 8 (0x4a5250).
- Id 1: the same field layout for **one** server, appended to the current list (form field +0x330). If it is then the only row, it is selected. This is an "add one server" message, not a login answer.
- So the lab's single frame (id 0 list, id 2 status) and the earlier two-frame 2.17-style reply are both accepted.

Connect [bin]

1. Connect button (0x4a604c): needs non-empty account and password, keeps them, then calls 0x4a61e0.
2. 0x4a61e0: the selected server must have at least 8 fields. **Field 6 is the host and field 7 is the port**, copied to the globals at 0x5397a4 and 0x5397a8. Fields 8–15 are optional (two strings and two numbers when there are at least 12, the same again for 16). Then 0x52a458 and **0x525c0c** run, and the game form is shown (0x44aa60).
3. 0x525c0c: **returns at once if the byte at 0x53a610 is set.** Otherwise it lays out the game window, then creates `TPlayer` (0x50ca0c).
4. `TPlayer` constructor: loads `letters.png` and `head26.gif`, does `SetCurrentDirectory(<game dir>levels)`, and when the host global is non-empty creates `TAdventureClient` (0x4dbc40).
5. 0x4dbc40: if the host or port is empty it shows "The connection could not be established!" without dialling. Otherwise it calls 0x4db17c: `WSAStartup` once, a name cache (0x4dad94), `inet_addr`, then `gethostbyname`, `socket`, `TCP_NODELAY`, `connect` to `htons(StrToInt(port))`, non-blocking. Any failure raises, and the same message is shown.

The lab list (name, language, description, url, version, players, ip, port) puts the IP and port where 2.00 reads them.

The fatal flag

0x53a610 is set only on a fatal error, right after `Application.MessageBox` and before `Application.Terminate` (0x44da90, which is `PostQuitMessage`). It is never cleared. The message is "File not found" / "The file X has not been found in the Graal folder!". **[bin]** The sites are in the asset loader 0x52525c (`sprites.png`, `pics1.png`, `state.png`, `opps.png`, `bomb1.png`, `bomb2.png`, and `objects.dat` via 0x52d36c → 0x52d3c1 `FileExists`), plus `arrays.dat` (0x52330d) and some quit paths.

0x52525c runs from the main menu's Start handler (0x4a78d0), before the server list opens. With `objects.dat` missing, the flag is set there. The list window still opens and works, but Connect stops at step 3. In the lab no dialog was seen after Connect. **[cap]** Whether the "objects.dat" box appeared at Start, or was hidden behind the menu, was not recorded.

`head26.gif` was also missing from the lab folder, but `TPlayer` loads it through `TSurfaceOwner` (VMT 0x4f3ca0, load at slot +8, 0x4f3d28), which has its own exception handler. It is not fatal.

Lab rule for 2.00

Run 2.00 from a folder that has its own `objects.dat` (the 2.22 distro has none), plus `arrays.dat`, `letters.png`, `sprites.png`, `pics1.png`, `state.png`, `opps.png`, `bomb1.png` and `bomb2.png`. The earlier lab run that reached the gserver used the 2.22 folder plus an `objects.dat` (`decomp-findings-2026-10-24-b/07-lab-findings-46-186-200.md`).

Confirmed [cap], 2026-10-30: after `objects.dat` from NewMain 26 (2001-01-15, 33852 bytes) was copied into the same folder, a re-run with the same lab servers showed no dialog at Start or Connect. 2.00 opened TCP to the gserver on 14802, drew the server's start level and chatted. Nothing else in the folder or on the servers changed.

`client-internals/version-id-map.md`

Version ID map

Date: 2026-10-14

Protocol token → version

GNW tokens are `GNW` + build date as DDMMY (GNW03081 = 3 Aug 2001). The 2.1 test builds (Sep 2001) still send GNW03081. Per-build hashes, dates and the rest are in `build-catalogue.md`.

Token	Version	Source
`GNW07080`	1.39 (CLVER_1_39)	version-token pack / gs2lib
`GNW13110`	1.41r1	allowedversions.txt
`GNW03081`	2.05 rev 2 / "2002ver9" leak (CLVER_2_052, sorts below CLVER_2_1)	gs2lib 2026; GServer versions.txt; built char-by-char in the 2.05 login
`GNW24011`	2.0 (and rev1–rev3)	version-token pack; confirmed in all four 2.00 exes (2026-10-24), see `version-tokens.md`
`GSERV007`	1.21	build catalogue
`GSERV013`	1.28 beta2	build catalogue
`GSERV014`	1.3 test	build catalogue
`GSERV016`	1.31	build catalogue
`GSERV018`	1.32	build catalogue
`GNW20020`	1.33	build catalogue
`GNW22030`	1.34	build catalogue
`GNW13040`	1.35	build catalogue
`GNW10050`	1.36	build catalogue
`GNW21050`	1.37	build catalogue
`GNW22060`	1.38	build catalogue
`GNW08011`	NewMain client (1.42 notes)	build catalogue
`GNW23021`	2.01, 2.01 rev1	build catalogue
`GNW15041`	2.02, 2.02 rev1	build catalogue
`GNW20041`	2.02 rev2	build catalogue
`GNW06051`	2.03	build catalogue
`GNW24061`	2.04	build catalogue
`G2N15041`	NewWorld betas 1 and 2	build catalogue

Token	Version	Source
`GNW22122`	2.1.7.0–2.1.7.1 (CLVER_2_17)	allowedversions.txt; gs2lib; GServer versions.txt
`GNW03014`	2.2.2.0 (CLVER_2_22)	allowedversions.txt; gs2lib
`GNW28015`	2.3.1.0 (CLVER_2_31)	allowedversions.txt; gs2lib
`GNW30123`	listserver token of the 2.22 and 2.31 clients (sent in listserver id 7 as key + `GNW30123newmain`), not a game build	2.22/2.31 binaries; seen on listserver port 14922; 2.22 confirmed in the lab (`protocol/listserver.md`)
`G3D03014`	Linux 3.04 rev 1 (= 3.041 in the server table below)	3.x client binary (same date stamp as GNW03014)
`G3D28015`	Mac OS X 3.153	3.x client binary (same date stamp as GNW28015)
`G3D30123`	3.x listserver login	3.x client binary
`G3D17026`	Windows 4.04r2	4.x engine DLL (date stamp 17 Feb 2006)
`G3D27076`	Mac OS X 4.20r6	4.x client binary (date stamp 27 Jul 2006)
`G3D20126`	Windows 4.20r8	4.x engine DLL (date stamp 20 Dec 2006)

Filename → build

Binary	Build
`Graal_2171.exe`	2.171
`Graal_222.exe`	2.22
`Graal_231.exe`	2.31

Construction (v2.05)

Token is eight `_LStrFromChar` immediates in `TAdventureClient_Create`, not a string literal. Random prefix `(rand & 0x7F) + 0x20` precedes it.

Update 2026-10-21: the login line is GCHAR client type 0, then GCHAR key (`rand & 0x7F`), then the token. The "random prefix" is that key (see `protocol/login-and-walk.md`).

Confirmed in Graal 139.exe.c

The login/token builder builds the token as char immediates: G N W 0 7 0 8 0 → GNW07080. Connects to port 14900 (PTR_s_14900).

1st Graal.exe.c

No GNW token construction found. Uses .zelda level files (level33.zelda, house19.zelda). Has socket code but no version-token immediates. Predates the GNW scheme.

Dhrake-improved 1.39 export

Same binary, better symbols. ~390 recovered names. Token still built as char immediates (line ~128320). Preferred corpus in shdecompiling/graal-139-dhrake.

Full gs2lib client token table (2026-10)

Token	Build	Token	Build
`GSERV007`	1.21	`GNW20052`	2.151

Token	Build	Token	Build
GSERV009	1.25	GNW12102	2.16
GSERV011	1.27	GNW22122	2.17
GSERV013	1.28	GNW21033	2.18
GSERV014	1.30	GNW15053	2.19 (bz2 + new encryption)
GSERV016	1.31	GNW28063	2.2
GNW07080	1.39	GNW01113	2.21 (compression byte)
GNW12080	1.391	GNW03014	2.22
GNW13110	1.41r1	GNW14015	2.3
GNW03081	2.05 rev 2	GNW28015	2.31
GNW31101	2.1 (2.1.0.0–2.1.0.3, also 2002ver10/13)	G3D16053	3.0
GNW01012	2.12	G3D27063	3.01
GNW23012	2.13	G3D03014	3.041
GNW30042	2.14	G3D28095…	4.x+
GNW19052	2.15		

The older GServer-v2 submodule lists 1.33–1.38 as `GNW20020`, `GNW22030`, `GNW13040`, `GNW10050`, `GNW21050`, and `GNW22060`. Its `versions.txt` writes them without the `G` (`NW20020`, …).

`client-internals/version-profiles/list-ports.md`

Default listserver endpoint per profile

Date: 2026-10-25 (updated after the login-type matrix)

This is what a reimplementation should dial first for each profile. It is **not** what the binary carries. The binary's own host and port are kept as history in `built_in_host` and `built_in_port`.

Rule (written in Seihai `documentation/port-rule.md`; evidence in `../../protocol/listserver-login-type-matrix-2026-10-25.md`):

- Host is always `<list-host-reborn>`.
- **14921 is dead.** The public service resets it. No profile defaults to it.
- The public listserver answers by **login type**, not by version or token. A classic login (one plain zlib frame, id 0 then id 1) gets a list on **14923** and nothing on 14922. A gen-5 login (id 7 with `GNW30123`) gets a list on 14922 and on 14923.
- **Legacy** profiles send the classic login and default to **14923**: 1.41 to 2.17, including 2.05 (Seihai's 2.05 profile sends the classic login).
- **Supported** profiles send the gen-5 login and default to **14922**: 2.22, 2.3, 2.31.
- `kind` is `legacy` or `supported`. `port_status` is `confirmed` (a public check of Seihai's login for that profile got a list on that port; a real client of that version was not captured) or `assumed` (not run with that login).

The table is produced by `python3 list_ports.py` (this directory) from the three profile files. `python3 list_ports.py --check` fails if a profile breaks the rule, so a later edit to a TOML cannot quietly bring 14921 back.

version	host	port	kind	port status	built-in host	built-in port
1.33	-	-	no listserver	-	-	-
1.34	-	-	no listserver	-	-	-

version	host	port	kind	port status	built-in host	built-in port
1.35	-	-	no listserver	-	-	-
1.36	-	-	no listserver	-	-	-
1.37	-	-	no listserver	-	-	-
1.38	-	-	no listserver	-	-	-
1.39	-	-	no listserver	-	-	-
1.41	<list-host-reborn>	14923	legacy	confirmed	<list-host-reborn>	14923
1.42	<list-host-reborn>	14923	legacy	confirmed	<list-host-classic>	14921
2.00	<list-host-reborn>	14923	legacy	confirmed	<list-host-classic>	14921
2.01	<list-host-reborn>	14923	legacy	confirmed	<list-host-classic>	14921
2.02	<list-host-reborn>	14923	legacy	confirmed	<list-host-classic>	14921
2.02 rev2	<list-host-reborn>	14923	legacy	confirmed	<list-host-classic>	14921
2.03	<list-host-reborn>	14923	legacy	confirmed	<list-host-classic>	14921
2.04	<list-host-reborn>	14923	legacy	confirmed	<list-host-classic>	14921
2.05	<list-host-reborn>	14923	legacy	assumed	<list-host-classic>	14921
2.1	<list-host-reborn>	14923	legacy	confirmed	<list-host-classic>	14922
2.12	<list-host-reborn>	14923	legacy	confirmed	<list-host-classic>	14922
2.13	<list-host-reborn>	14923	legacy	confirmed	<list-host-classic>	14922
2.14	<list-host-reborn>	14923	legacy	confirmed	<list-host-classic>	14922
2.15	<list-host-reborn>	14923	legacy	confirmed	<list-host-classic>	14922
2.16	<list-host-reborn>	14923	legacy	confirmed	<list-host-classic>	14922
2.17	<list-host-reborn>	14923	legacy	confirmed	-	-
2.22	<list-host-reborn>	14922	supported	confirmed	-	-
2.3	<list-host-reborn>	14922	supported	confirmed	-	-
2.31	<list-host-reborn>	14922	supported	confirmed	-	-
3.04	<list-host-reborn>	14922	supported	assumed	<list-host-3x>	14922
3.153	<list-host-reborn>	14922	supported	assumed	<list-host-3x>	14922

What the status means per profile (matrix of 2026-10-25, one list-only login per profile and port):

status	profiles	evidence
confirmed	1.41, 1.42, 2.00 to 2.04 (with 2.02 rev 2), 2.1, 2.12 to 2.16, 2.17	Seihai's classic login got a legacy list on 14923 (2 or 4 rows) and no reply for about 5.5 s on 14922, in every case. Real clients of these builds were not captured; 2.1 to 2.16 have no binary at all
confirmed	2.22, 2.3, 2.31	the gen-5 login got the modern list on 14922 (8 rows, welcome, site and upgrade lines) and on 14923 (12 rows, 2.31: 17)
assumed	2.05	Seihai's 2.05 profile sends the classic login, which was not run for 2.05. The matrix binary sent the gen-5 login for 2.05; both ports answered it. 14923 answered the classic login of every other classic profile, so it is the port assumed here
assumed	3.04, 3.153	the 3.x clients send the gen-5 login with `G3D30123newmain` (`../../version-profiles/3.04-3.153.toml`). A G3D token was not tried against the public listserver; 14922 is assumed from the 2.22/2.31 result

Open: a classic 2.05 login on 14923; how a real 2.17 client logs in to the listserver.

`client-internals/version-profiles/README.md`

Version profiles, 1.41 to 2.17

Date: 2026-10-23

`141-217.toml` describes each client release from 1.41 rev1 to 2.17 well enough for a reimplementation to *be* that version. That covers what it sends and understands on the wire, and also what the player sees and can do. `resolve.py` flattens the inheritance (`base`, `_add`, `_remove`) and prints JSON or one markdown block per version, with a source tag on every value:

```
python3 resolve.py 141-217.toml          # JSON
python3 resolve.py 141-217.toml --md     # markdown
```

Field names follow the Seihai profile schema: `token`, `listserver`, `client_type`, `generation`, `login_key_byte`, `compression`, `cipher`, `pli_sent`, `plo_known`, `props_max`, `prop_diffs`, `login_props`, `level_formats`, `file`, `tampercheck`, `udp`, and `features`.

Sources

Tag	Meaning
bin	Read from that build: switch tables, send sites, DFM forms, strings, shipped images. 24 executables were scanned (1.41 rev1, NewMain 24/26, 2.0 r0–r3, 2.01 r0–r1, 2.02 r0–r2, 2.03, 2.04, 2.05 r0–r2, 2.1 test 2/3, NewWorld betas 1/2, 2.1.7.1)
doc	The release notes shipped with the clients (`newfeatures141/142/2001/2002.txt`)
srv	GServer-v2 source: what a server does differently per version
forms	The DFM resources: start window, options, controls, server list, player list
lab	Seen on the wire from the real client, run under Wine against a lab list and game server (2026-10-30: 1.41 rev1, NewMain 26, 2.00 rev0, 2.02 rev1, 2.02 rev2, 2.05 rev2, 2.1 test 3, NewWorld betas 1/2, 2.1.7.1). A lab tag covers only the build that was run; `resolve.py` turns it into `inf` in the profiles that inherit it
inf	Not checked in that build. Either the value is carried from the nearest scanned build, or it is placed at the release that introduced the feature

No binaries of 2.1 release, 2.12, 2.13, 2.14, 2.15 or 2.16 are in the archive. Their entries are release notes plus server source, and every value they inherit from a scanned build is tagged `inf`.

How the packet sets were read

- **Server → client:** the receive dispatcher is a `case` on `id` with a jump table. The ids with a non-default arm are the ids the build handles. The results match `cross-version-timeline.md` except for one correction (below).
- **Client → server:** every packet goes out through one virtual send method that takes the id as a byte argument. Scanning each build for constant ids passed to that call reproduces the known 2.05 set exactly (`packet-ids.md`), so the same scan was used for the other builds. Id 0 is sent through a computed path and is present in all of them.
- **Props:** each prop parser and the prop writer are `case` tables too. The writer's table is the list of props the client can send. In 2.05 it matches the "sent by the client" list in `player-props.md`.

Wire boundaries

Change	First build	Tag
gen 3 (key byte + LCG insertion), server list window; the listserver speaks first and the client sends nothing	1.41	bin, lab
Client asks the listserver for a list by name (id 0 `newmain`), no account	1.42	lab
`.gani`, light effects, PLO 42, 44, 45, 150, 151, 153; **UPDATEFILE (34) sent**	1.42	bin
`.nw` loader dropped (until 2.05; GraalX keeps it), listserver host DES-hidden, PLO 154, 155	2.00	bin
ADJACENTLEVEL (35), HITOBJECTS (36) sent; PLO 46, 156	2.01	bin
PLO 47; player props 37–41 (`#P1`–`#P5`) parsed and sent; NPC props to 40	**2.02**	bin
LANGUAGE (37) sent after login	2.02 rev2	bin, lab
TRIGGERACTION (38) sent; PLO 48, 165	2.03	bin
PLO 166, PLO 26 dropped; prop 42	2.04	bin
`.nw` + `.gmap`; PLO 168	2.05	bin
Account and password sent to the listserver (id 1 after the id 0 list name); NewWorld beta 1, on the 2.05 code, sends only the list name	2.1 test 3	lab
Bow slot → GANI string, `.png` defaults, PLO_FILE modtime, FILEUPTODATE, `sprite % 4` for NPCs, prop caps lifted	2.1 (GNW31101)	srv
Large files (LARGEFILESTART/END)	2.14	srv, doc
TAMPERCHECK (39), SHOOT (40), SERVERWARP (41), MUTEPLAYER (43), PROCESSLIST (44/45); signed `arrays2.dat` listserver	2.17 (checked); some may be earlier	bin
Login props add PLANGUAGE (52); 39 and a PLI 44 window-title list are sent at login	2.17 (checked)	lab

Corrections to earlier notes:

- **PLO 47 and props 37–41 are already in 2.02**, not first in 2.02 rev2 as `cross-version-timeline.md` says. That note skipped the 2.02 base build.
- **1.41 has no UPDATEFILE sender.** It never revalidates a cached file. 1.42 adds it, which matches the 2.00 note "files are now automatically downloaded again from the server when they have been updated".
- The 2.02 rev1 notes say the client language is sent to the server. No LANGUAGE send site exists in rev1, and the lab run confirms it sends none. The first one is in rev2.
- The 2.1 test builds still send 2.05's token (GNW03081), so a server cannot tell them apart from 2.05 and uses the pre-2.1 paths for them.

What the player sees

These are the `features` keys, which matter most for making the client look and behave like the chosen version.

- **1.41.** The start window lists game modes (One Player, One Player Online, Two/Three/Four Players) beside character heads. Local 2–4 player split screen and save games (F2/F3) still exist. There are no ganis, so animation runs on sprite numbers only. The HUD gets its AP bar under the ASD box this version, plus a "sparring" label and the R key for ratings. Right-clicking a player opens their profile. The options window

has a Save games page. The HUD sheet has two chat-bubble colours and no gralat bag, clock or day/night icons.

- **1.42 / 2.00.** The game becomes online-only. The start window shows a character with Training (built-in tutorial levels) and Start buttons. The 2.00 start window is titled "Graal 2001". Keys are rebindable in the Controls window. Ganis and light effects arrive, with a "Don't use DirectX7" option. The HUD sheet adds admin, guild and mass chat-bubble colours, a gralat bag, a clock and day/night icons. The player list gets a "Staff" section (from 2.00).
- **2.01–2.04.** Levels preload on the map, and a map-reading animation plays. 2.02 adds hats (`#P1`–`#P5`), images in signs and HTML-window navigation. 2.04 adds `replacegani`, tile replacement, attaching the player to an object and a weapon-select toggle.
- **2.05.** The server name appears in the title bar. `.nw` and `.gmap` levels load. The shipped HUD sheet adds the "PK off" sword icon (identical to the 2.1 test sheet), although the notes date that feature to 2.1.
- **2.1.** A new start window and server list show account info. The options window gets tabs (General, Video, Audio, Controls) and a Direct3D toggle. Ghost mode, auto-mapping, map and minimap set by the gserver, the Alt+0 log, smilies in PMs and gani scripts all arrive, and default images switch from GIF to PNG/MNG.
- **2.12–2.16.** The editor gains multi-level editing (2.12). Gani attributes grow to `#P9` (2.13) and then `#P30` (2.16). The server list splits into two tabs (2.14), which also bring voice chat, OGG/MP3/MOD music, volume sliders, player status and `serverwarp`. 2.16 adds transparency modes, `enablefeatures` and `showani`/`showpoly`/`showtext`.
- **2.17.** The start window gains payment buttons and a "Private Server List". The options window gains Audio and Microphone tabs and an Automapping toggle. The player list gets status (Online/Away/DND) and Mute. The client handles the RPG window, Kingdoms and New World modes, and level heights.

Lab runs (2026-10-30)

Each client was run under Wine with its own data folder against a stub list and game server. The stub answers WANTFILE with FILESENDFAILED. Per build:

Build	Listserver	Login line	First messages after login	Login props
1.41 rev1	14923; server sends the list first, client sends nothing	type 0, GNW13110, account, password	none	0, 11, 13, 18, 31, 33
NewMain 26 (1.42)	14921; id 0 `newmain`	type 0, GNW08011	file requests	same
2.00 rev0	same as 1.42	type 0, GNW24011	file requests	same
2.02 rev1	same as 1.42	type 0, GNW15041	file requests (no LANGUAGE)	same
2.02 rev2	same as 1.42	type 0, GNW20041	LANGUAGE, file requests	same
2.05 rev2	same as 1.42	type 0, GNW03081	LANGUAGE, file requests	same
2.1 test 3	14922; id 0 `newmain`, then id 1 account + password	type 0, GNW03081	LANGUAGE, file requests	same
2.1.7.1	same as 2.1 test 3	type 0, GNW22122	LANGUAGE, TAMPERCHECK, file requests, PLI 44	adds 52; head as a file name

The ports are the ones built into each exe (the lab routed every list host to itself). Every login line is the same layout as in `defaults`. Head (11) is an index below 100 in every build up to 2.1 test 3.

Notes for a server or a reimplementation:

- 1.41 stalled at "Loading account from server (1/5)" when the login burst carried 2.05-era ids it does not handle. With only ids from its `plo_known` it entered the game.

- 2.00 waits for `idle.gani` before it sends its login props. If the file is missing locally and the server cannot send it, the client sits in the level with no body.
- 2.02 rev1 and rev2 load `zlib.dll` (cdecl). The archive copies ship without it, so a lab needs one built from zlib.
- Under Wine, the 2.x clients sometimes stopped sending after the login burst when the in-game HTML window opened at the same moment. A rerun always went through, so this is put down to the lab, not the clients.

Open

- 2.1–2.16 packet and prop sets are inferred from neighbours. A binary of any of them would pin them down.
- The original 1.41 listserver host and port: the only surviving 1.41 copy is patched for <graalin-site>.
- Login props were not run for 2.01, 2.03 or 2.04. They are inferred from their neighbours, which all send 0, 11, 13, 18, 31, 33.
- What 2.17's periodic PLI 24 (one byte, every 5 s) carries.
- Which 2.05-era server id stalls 1.41 at "Loading account from server (1/5)".
- The 2.17 other-player prop parser and its GS1 command count were not read.

Default list port

`listserver` carries the endpoint a reimplementation should dial first: `host`, `port`, `kind` (`legacy` or `supported`) and `port_status` (`confirmed` or `assumed`). It is not the value the binary carries: that is kept as history in `built_in_host` and `built_in_port`.

- Host is always `<list-host-reborn>`.
- 14921 is dead (the public service resets it) and is nobody's default.
- The default follows the login type the profile sends (the public listserver answers by login type, not by version or token): legacy = classic login (1.41 to 2.17, including 2.05) defaults to 14923; supported = gen-5 login (2.22, 2.3, 2.31) defaults to 14922.
- Every profile is `confirmed` except 2.05: Seihai's own login for it was run on its port in the matrix of 2026-10-25 (a real client was not captured). 2.05 is `assumed`: its classic login was not run.

The per-profile table and the evidence behind each status are in `list-ports.md`. `python3 list_ports.py --check` fails when a profile breaks the rule. Rule text: Seihai `documentation/port-rule.md`; public results: `../../protocol/public-listserver-2026-10-25.md` and `../../protocol/listserver-login-type-matrix-2026-10-25.md`.

1998 to 1.32 (added 2026-10-30)

Two more files cover the clients before 1.33. They use the flat layout of `v133-v139.toml`: each `[[version]]` holds every value, with no `base`.

- `v1998-v11.toml`: 20 builds from 1st Graal (Sept 1998) to 1.1 (Feb 1999). Generation 1, the text relay in `../../protocol/relay-1998.md`. There is no token, login or account, so `token` and `pli_sent` are replaced by `relay_game` (the game name sent with `N`) and `d_types_sent` (the relay `D` type letters). It also gives the dialog host, the level and save magics, and the config keys.
- `v121-v132.toml`: 1.21, 1.28 beta 2, 1.3 test, 1.31 and 1.32. Generation 2, GSERV tokens, see `../../protocol/gserv-1999.md`. Same keys as the 1.33 file, plus `plprops_sent` (the prop writer's table) and `bundled_servers`.

How the gen-2 sets were read: the same switch-table and send-site method as above, with each table found in 1.32 and then matched in the older builds by function shape. `pli_sent` is a lower bound here. 1.28 beta 2 and 1.3 test pass more ids in a register, which is why they list fewer ids than 1.21.

	1.21	**1.28 b2**	**1.3 test**	**1.31**	**1.32**
Server ids	0–34 except 30, 100	0–38, 100	0–39, 100	0–40, 100	0–41, 100
Player props parsed	0–24	0–31	0–32	0–32	0–33
Player props sent	0–24	0–24, 26, 31	0–24, 26, 31, 32	same as 1.3	0–24, 26, 31–33
NPC props	0–17	0–32	0–34	0–34	0–34
UDP ids	none	0, 10–22	0, 10–22	0, 10–23	0, 10–23
GS1 commands	71	84	85	85	85

1.32's sets match the 1.33 profile, except that 1.33 adds three GS1 commands (`setbody`, `setfocus`, `resetfocus`).

Correction to `v133-v139.toml`: `save_format` said `Graalsav9`. These builds write `Graalsv11`. The three save magics are referenced the same way as in 1.32, where the writer was traced (see `../../formats/graalsav.md`).

No build in this range was run. Every value comes from the binaries and the shipped notes. The features tagged `doc` come only from the release notes.

`client-internals/version-tokens.md`

Version tokens

Date: 2026-10-14

Internal version strings recovered from the public version list. No decompile material.

Token	**Build notes**
GNW07080	1.39
GNW03081	2.05 rev 2 (= 2002ver9 leak build); gen-3 cipher, pre-2.1 code paths
GNW31101	2.1.0.0–2.1.0.3 (version string not updated)
GNW30042	2.1.4.0 (first with server-list tabs)
GNW15053	2.1.9.0 (bz2 + new encryption, new server-list gfx)
GNW01113	2.2.1.1 (plaintext and zlib; extra compression byte)
GNW14015	2.3.0.0
GNW28015	2.3.1.0
G3D03014	Linux 3.04 (Torque rewrite; 2.22 protocol, see `protocol/v3x-clients.md`)
G3D28015	Mac OS X 3.153 (2.31 protocol)
G3D30123	3.x listserver token (the 3.x form of GNW30123)
G3D17026	Windows 4.04r2 (`GraalEngine.dll`, March 2006)
G3D27076	Mac OS X 4.20r6 (Delphi launcher + C++ engine era; see `protocol/v4x-clients.md`)
G3D20126	Windows 4.20r8 (`GraalEngine.dll`)
G3D20126	Windows 4.20r8 (engine DLL; logs into the `Login` game server)
G3D22067	Windows 5.00r7 (engine DLL, Jul 2007; see `protocol/v5x-clients.md`)
G3D14097	Windows 5.12 (lab capture; see `protocol/v5v6-windows-unpacked.md`)
G3D03048	Windows 6.00 beta1 (lab capture)
G3D15010	Windows 6.00 beta2, 5.24r3, 5.24r4 (lab capture)
G3D26090	Windows 5.30r3, 5.31r4, 6.00r2, 6.00r4 (lab capture)
G3D0311C	Windows 6.03r7 (lab capture; Linux 6.03r7 sends G3D0511C)

Token	Build notes
G3D0108A	Linux 6.01r1 (see `protocol/v6x-clients.md`)
G3D0511C	Linux 6.03r7

Filename builds in the client tree: Graal_2171 = 2.171, Graal_222 = 2.22, Graal_231 = 2.31.

From 2.3 the editor and the online client are separate executables. PE FileVersion is authoritative for the editor (2.3.1.1).

The early-client transition (14801 → 14900 → length+zlib) is the protocol-side counterpart of the same era.

Confirmed from the binaries (2026-10-24)

The 2.0x family tokens below were read from the character immediates of each client's login builder, so they are confirmed per build rather than taken from a token list. Every value matches the earlier tables here and in `version-id-map.md`. Machine-readable copy: `version-profiles/version-tokens-by-build.tsv`.

Token	Build	Source
GNW24011	2.00 and rev1, rev2, rev3 (all four exes)	login builder
GNW23021	2.01, 2.01 rev1	login builder
GNW15041	2.02, 2.02 rev1	login builder
GNW20041	2.02 rev2	login builder
GNW06051	2.03	login builder
GNW24061	2.04	login builder
GNW03081	2.05, rev1, rev2; also the 2.1 test builds (`graal21test2`, `graal21test3`), which still send the 2.05 token	login builder
GNW08011	NewMain 24 and 26	login builder
GNW13110	1.411	login builder
GNW07080	1.39	login builder
GNW22060	1.38	login builder
GNW22122	2.17.1	lab capture and exe
GNW03014	2.22 game login	exe
GNW28015	2.31 game login	lab capture
GNW30123 + `newmain`	2.22 and 2.31 listserver hello (not a game build)	exe and lab

A lab GServer-v2 only needs the token in its allowed-versions list to let a client in. For 2.00 (`GNW24011`) that is a config entry, not a code patch (`client-internals/decomp-findings-2026-10-24-b/04-v2.00-client.md`).

`client-internals/version/docs/GraalEditor-architecture.md`

Architecture

GraalEditor is a Delphi VCL application. The live document is a `TAdventureLevel` owned by the main form `TAdventure`. Preview goes through DelphiX (`TDXForm`, DirectDraw 5 and 7). Sound goes through BASS. Picture codecs are the same family GraalShop uses (GIF, MNG, PNG/DIB).

```
GraalEditor.exe  (unpacked image, base 0x400000)
├── Delphi RTL          AnsiString, exceptions, file IO
```

```
├── VCL                 forms, listviews, richedit, menus, dialogs
├── Graphics            TGIF, TMNG, TJPEGImage, TBitmap, TGraalImage
├── DelphiX / DirectDraw  TDXForm, DirectDraw 5 and DirectDraw 7
├── BASS                music, samples, microphone device names
├── Domain              TAdventureLevel, TEditorLevelCache, TGraalMap,
│                       TEditorLink, TEditorSign, TEditorNPC, TEditorChest,
│                       TEditorObject, TLevelObject, TLevelHeights
├── Animation           TGraalAni and the GraalShop actor/step/sprite set
└── UI                  TAdventure + modal frames (see ui.md)
```

See diagrams/architecture.svg and diagrams/classes.svg.

Layers

Runtime

Borland RTL banner: `SOFTWARE\Borland\Delphi\RTL`, `Portions Copyright (c) 1983,97 Borland`. Linker version 2.25. Sections are the classic Delphi layout: `CODE`, `DATA`, `BSS`, `.idata`, `.tls`, `.reloc`, `.rsrc`. Unit name recovered from RTTI: `UAdventureEditor`.

VCL shell

One main form, twenty modal or tool forms stored as `TPF0` DFMs in `.rsrc`. The main form is `TAdventure` / `Adventure`, caption `Graal`, `poScreenCenter`, `KeyPreview` set. It owns the tile canvas, a mode radio group, and the menu that opens the other frames.

There is no plugin ABI. Registry use that shows up is the Delphi locale key and the shell-folders key the Thinstall stub virtualizes, not a Graal account store. Account files the editor knows by name: `buddyaccounts.txt`, `ignoredaccounts.txt`, `mutedaccounts.txt`, `graalconfig.txt`, `headsconfig.txt`, `FILENAMECACHE.txt`.

Picture, map, sound

- GIF: `TGifAnimation`, `TGifAnimationStep`, failure `Gif loading failed!`
- MNG: `TMNGAnimation`, `TMNGAnimationStep`, failure `MNG loading failed!`
- PNG and ordinary bitmaps via `TGraalImage` / `TBitmap`. Image dialog: `CImage files (*.png, *.mng, *.gif)`
- DirectDraw init can fail for 5 or 7: `Error when loading and initializing DirectDraw5` / `DirectDraw7`. More Options has `Don't use Direct3D (lights)` and `Draw in compatibility mode (very slow)`, plus a vertical-blank wait checkbox.
- BASS loads music and samples. More Options has Audio and Microphone pages.
- World map assets: `graalmap.png`, `map.png`, `map2.png`, `minimap.txt`, `bigmap.txt`, `loadgmaps.txt`. Status strings: `Loaded gmap`, `Loaded minimap`, `Loaded minimap from`.

Domain

`TAdventureLevel` is the open level. `TEditorLevelCache` keeps recently loaded levels (the link walker and the gmap need neighbours). `TGraalMap` is the stitched world. `TLevelHeights` is a height overlay distinct from the tile board.

Placed entities are separate types, not one tagged union: `TEditorLink`, `TEditorSign`, `TEditorNPC`, `TEditorDefNPC`, `TEditorChest`, `TEditorObject`, plus baddy state edited by `TEditBaddy`. `TEditorUndoInfo` is the undo record. `TGaniObject` / `TImageAnimation` attach a GANI to a placed object.

Animation subset

The GANI object model is the GraalShop one, compiled into the editor so tile-object scripts and NPC animations can be previewed without launching GraalShop:

`TGraalAni`, `TGraalAniSprite`, `TGraalSpriteAttachment`, `TGraalAniPart`, `TGraalAniSound`, `TGraalAniStep`, `TGraalActor`, `TGraalActorMod`.

`TTilesArrayFrame` is the tile-index / script editor (Courier New edit plus a rich edit). Its sample script is the offline tutorial: `addtiledef picso.png,offlinetutorial,1` and `setplayerprop` / `setcharprop` lines.

Control flow, one screen

```
WinMain-equivalent (Delphi RTL)
        │
        ▼
TAdventure  caption "Graal"
        │
        ├── F5 Load Level ── open *.graal / *.nw ── TAdventureLevel
        ├── F4 Edit Level ── tile / object / baddy / npc / chest radios
        ├── links ── TLinkList ── TLinkOptions
        ├── signs ── TEditSigns
        ├── NPC ── TEditNPC (image + script rich edit)
        ├── baddy ── TEditBaddy (type + three verses)
        ├── chest ── TChestEdit (item + sign index)
        ├── F6 Debugger, F7 Player List, F8 Server List
        └── save ── write BOARD / LINK / SIGN / CHEST / NPC / BADDY
```

Binary facts

- Unpacked image base `0x400000`, `SizeOfImage 0x1B1000`, entry RVA `0x155D4C`.
- Version resource (also present, UTF-16, in the packed stub): CompanyName `Linux Cyberjoueur`, FileVersion `2.3.1.1`, InternalName `Graal editor`, ProductName `GraalEditor`.
- On-disk packed file is the Thinstall stub. Do not point IDR or a Delphi decompiler at `graal/GraalEditor.exe`. Point them at `unpacked/GraalEditor.exe`.

`client-internals/version/docs/GraalEditor-README.md`

GraalEditor.exe — reverse-engineering notes

GraalEditor 2.3.1.1 is the Graal Online classic level editor (Linux Cyberjoueur). The file in `graal/GraalEditor.exe` is not a Delphi image. It is a Thinstall (pre-VMware ThinApp) package. That is why IDR rejected it, and why the Ghidra C dump in `Export/GraalEditor.exe.c` is only 3,136 lines of loader.

The Delphi editor was recovered by running the stub decompressor and then the Thinstall `0xD7` block codec. The unpacked image is `unpacked/GraalEditor.exe` (1,729,536 bytes, Borland linker 2.25). Names below come from Delphi RTTI and binary DFM resources (`TPF0`), not from original source.

Why the dump looks empty

`Export/GraalEditor.exe.c` decompiles the outer stub only (entry into the stub startup, then the package-open helper). The optional-header `LoaderFlags` field is reused as a file offset (`0xFC94E`) of a 0x38-byte package header. Magic is `0x82D0B188`. The first block is the Thinstall runtime (`X:\thinstall\stub\ts_stub.cc`), not the editor. Ghidra never saw the Delphi code, so IDR had nothing with a Delphi VMT to reconstruct.

What the program is

A single-window level editor (`TAdventure`, caption `Graal`) with modal tools for links, signs, NPCs, baddies, and chests.

- Levels open and save as `*.graal` and `*.nw`. Filter string: `Graal levels (*.graal, *.nw)|*.graal; *.nw`.
- A level is a text file with `BOARD`, `LINK`, `SIGN`, `CHEST`, `NPC`, `BADDY` / `BADDYEND`, and `LEVELNAMES` / `LEVELNAMESEND` sections. `BOARD 0` is the tile layer.
- World view is a gmap (`*.gmap`, `loadgmaps.txt`, `bigmap.txt`, `minimap.txt`, `graalmap.png`).
- Preview is DelphiX DirectDraw. Both DirectDraw 5 and DirectDraw 7 init failures are stringed. Sound is BASS (`bass.dll` is imported by the stub's import table and by the editor).
- Spritesheets are PNG, GIF, MNG. Failures are `Gif loading failed!`, `MNG loading failed!`, `Image loading failed!`.
- The same GANI classes as GraalShop are linked in: `TGraalAni`, `TGraalAniSprite`, `TGraalAniStep`, `TGraalActor`, `TGraalActorMod`, plus `TGraalAniPart`, `TGraalAniSound`, `TGraalSpriteAttachment`.

Read this first

Document	Contents
packer.md	What the Ghidra dump actually is
architecture.md	Process, modules, class map
data-model.md	Level, link, sign, NPC, baddy, chest
level-format.md	On-disk sections the editor reads and writes
flows.md	Load, save, paint, mode switch
pseudocode.md	Reconstructed procedures
ui.md	Forms recovered from DFM
diagrams/	Packer, architecture, classes, load/save

Package contents

Thinstall virtual path `%InstallPath%`:

File	Role
`GraalEditor.exe`	Delphi editor, version resource 2.3.1.1
`hvdi.dll`	Bundled helper, 133,632 bytes. No Graal strings; not the editor core
`arrays.dat`	11,500-byte sorted dword table (count word `0x03A6`)
`arrays2.dat`	Server list: `<list-host-3x>:14922` and two fallbacks
`arrays2.sig`, `graaleditor.sig`	0x80-byte signatures. Stub string: `Not ready to verify signature, must have valid public key`
`thinstall.log`	Virtual log path

`Setup_Graal` is the package group name. `TS_EXECUTE_EXTERNAL` is set in the package directory.

`client-internals/version/README.md`

Seihai — Version ID Recovery

How the project recovers version identifiers from Graal classic binaries and their decompiles. Packaged for the Seihai project.

Contents

- `VERSION.md` — consolidated notes on the methods used.
- `docs/` — supporting documentation:
  - `GraalEditor-README.md`, `GraalEditor-architecture.md`
  - `graaleditor.txt`, `graal.txt` (LinuxMan 2 man pages)

Quick reference

Binary	Version signal	Value
GraalEditor.exe	PE version resource (FileVersion)	2.3.1.1
Graal_2171.exe	Filename	2.171
Graal_222.exe	Filename	2.22
Graal_231.exe	Filename	2.31
gif2mng.exe	Banner string	2004-06-03

The PE version resource is the reliable source for FileVersion / CompanyName / ProductName. It is present even in the Thinstall stub of GraalEditor. Client builds are identified by filename encoding.

`client-internals/version/VERSION.md`

Finding the Version ID from the Decompile

The version ID is recovered from the **PE version resource**, not from executable logic.

Primary method: PE version resource

In a Ghidra dump the structure appears as `VS_VERSION_INFO`:

```
struct VS_VERSION_INFO {
    word StructLength;
    word ValueLength;
    word StructType;
    wchar16 Info[16];
    byte Padding[2];
    dword Signature;
    word StructVersion[2];
    word FileVersion[4];      // four-part version
    word ProductVersion[4];
    dword FileFlagsMask[2];
    dword FileFlags;
    dword FileOS;
    dword FileType;
    dword FileSubtype;
```

```
    dword FileTimestamp;
};
```

The actual string values live in the `.rsrc` section as UTF-16. For GraalEditor the notes record them directly:

- CompanyName: `Linux Cyberjoueur`
- FileVersion: `2.3.1.1`
- InternalName: `Graal editor`
- ProductName: `GraalEditor`

That resource is present in both the packed Thinstall stub (`graal/GraalEditor.exe`) and the unpacked Delphi image. The packed-stub dump is only the loader (~3 k lines), so the version string is still readable from the stub's resources without unpacking.

Secondary signals

Source	What it gives	Example
Filename	Client build	`Graal_2171.exe` → 2.171, `Graal_222.exe` → 2.22, `Graal_231.exe` → 2.31
PE optional header	Linker version	Editor: Major/MinorLinkerVersion = 2.25
Banner / About strings	Date or product string	gif2mng: `"Graal Image conversion and palette fixing tool 2004-06-03"`; `TAboutDlg` caption "Version"
String search	Hard-coded tokens	`graalversion`, `Version`, product names

Practical steps on a new dump

1. Open the binary (or the Ghidra project) and locate the version resource under `.rsrc` / `VS_VERSION_INFO`.
2. Read the UTF-16 StringFileInfo block for `FileVersion`, `ProductVersion`, `CompanyName`, `InternalName`.
3. Cross-check the PE optional header for linker version if needed.
4. For the game clients, the filename itself is the version token used in the notes.

What the project recorded

- GraalEditor 2.3.1.1 (Linux Cyberjoueur). The file in `graal/` is a Thinstall package; the Delphi image was recovered by running the stub decompressor and the Thinstall `0xD7` block codec. Unpacked image is at `GraalEditor/unpacked/GraalEditor.exe` (Borland linker 2.25).
- Clients: `Graal_2171.exe` (2.171), `Graal_222.exe` (2.22), `Graal_231.exe` (2.31).
- gif2mng banner date: 2004-06-03.

All of the above is measured from the binaries and their resources. The version resource is the authoritative FileVersion; filenames supply the client build numbers used in the notes.

`client-internals/versions-133-139.md`

Client versions 1.33 to 1.39

Date: 2026-10-23, live check 2026-10-30

This note covers what makes each of the seven gen-2 GNW clients different, both on the wire and for the player. It is the writeup for `version-profiles/v133-v139.toml`, the machine-readable table that a client such as Seihai can load to behave as one exact build. The table gives every key in full for every version. This note only describes changes.

It is cleaned: no addresses, function names or decompiler output. The builds are the ones in `build-catalogue.md`, identified by MD5 there, and every shipped `Graal.exe` matched its catalogued hash.

Sources and method

What	How it was read	Confidence
Server → client ids, prop ids, UDP ids, GS1 command tables	Jump tables and the command table, as described in `cross-version-timeline.md`	confirmed
Client → server ids	Every call to the TCP send helper with a constant id. The helper is the one that prefixes `chr(id+32)`, appends `\n` and flushes at 4096 bytes.	confirmed for constant ids. A few call sites pass the id in a register and are not counted.
Login	The code that builds the first message after connect, read in 1.33 and 1.39	confirmed in 1.33 and 1.39. 1.34 to 1.38 are assumed to be the same.
Windows, options, menus	The Delphi form resources (captions and components)	confirmed
`graalconfig.txt` keys	Plain strings ending in `=`	confirmed. From 1.38 on some strings are built one character at a time, so those lists may be short.
HUD and tiles	`state.png`, `sprites.png` and `pics1.png` shipped beside each exe, compared tile by tile	confirmed
Player-facing behaviour	The shipped `newfeaturesNNN.txt` and `readme.txt`	confirmed as release notes, not checked in code
Server behaviour	GServer-v2 source (public). Its version checks only distinguish 1.41 and 2.1+, so it treats all of 1.33–1.39 the same. Checked live against GServer-v2 beta4, see "Live check".	confirmed live for 1.33 and 1.39
Live capture	1.33 and 1.39 run under Wine against a local GServer-v2 beta4, with the traffic captured and decoded. See "Live check" below.	confirmed for 1.33 and 1.39

What is the same in all seven

- **Wire.** Gen 2: every flush is one `u16be length` + zlib stream, with no size threshold, no cipher and no key byte. See `protocol/gserv-1999.md`.
- **Login.** Login is message id 0, which is `chr(32)` and also marks the client type as player. It carries the 8-byte token, then `chr(len+32)` + account, then `chr(len+32)` + password.
- **Server.** There is no listserver. The connect dialog is pre-filled with `<list-host-classic>:14900` and the `guest`/`guest` account. There is no server list window.
- **Levels.** Only GR-V1.00 to 1.03 load (`.graal`, plus `.zelda` Z3-V1 for the offline levels). There is no `.nw` and no `.gmap`. The board does arrive as `RAWDATA` (100, with an i3 length) followed by `BOARDPACKET` (101)

in its own frame (confirmed live). Saves are written as `Graalsv11` (the reader also takes `Graalsav9` and `Graalsv10`; see `../formats/graalsav.md`).

- **Files.** The client asks for files with `WANTFILE` (23) and never sends `UPDATEFILE` (34). It handles no large-file ids. The reply is `RAWDATA` (100, i3 length) then `PLO_FILE` (102): `chr(len+32)` + name, then the file bytes, with no modtime (confirmed live in 1.39). For clients before 2.1, the server adds `.gif` to image names that have no extension.
- **No tampercheck** (39).
- **UDP.** There is a peer UDP channel for "data packages", with `noudp=` and `fixedudpport=` (default 14899).
- **Game modes.** One Player, One Player Online, and Two, Three or Four Players on one machine. Local multiplayer goes away in 2.0.
- **Window.** The window is 640×480 (648×508 outer). Alt+Enter switches to fullscreen.
- **Keys.** F1 opens options, F2/F3 load and save, F4 opens the editor, F5 loads a level, F6 opens the debugger, F7 opens the player list, and Alt+1 to Alt+7 take screenshots, show the minimap, show the bytes counter, take a snapshot and record.
- **Prop encodings.** These are not traced per build. Live, the server-sent forms of props 1–9, 11 and 13 parsed in 1.33 and 1.39 as the pre-2.1 forms in `protocol/player-props.md`. The rest are assumed (*inferred*). The forms are: sword `c(power+30)+str`, shield `c(power+10)+str`, bow `c(10+len)+image` or `c(power)`, head `c(len+100)+image`, and X/Y in half-tiles. No build in this range has X2/Y2. Prop 10 must be the bow form; a server that sends a gani string there (GServer-v2 beta4 does) is misread, and the rest of that prop stream is lost.

Protocol changes

Build	Client → server	Server → client	Player props	Other-player props	NPC props	UDP
1.33	0–31	0–41, 100	0–33 *[inf]*	0, 2, 8–13, 15–21, 24, 30–32, 50, 51	0–34 *[inf]*	0, 10–23
1.34	same	same	0–33	+ 34	0–34	same
1.35–1.37	same	same	same	same	same	same
1.38	+ 32 `ITEMTAKE`, 33 `WEAPONADD`; − 25	+ 43 `DEFAULTWEAPON`	same	same	same	+ 24
1.39	+ 80 `PROFILEGET`, 81 `PROFILESET`	+ 75 `PROFILE`	+ 34 `ACCOUNTNAME`	+ 33	same	same

Some of these protocol steps line up with release notes.

- **1.34 adds other-player prop 34 (account name).** The same release saves ignores by account name (`ignoredaccounts.txt`).
- **1.38 adds `ITEMTAKE`.** The 1.38 notes say "when you get the item then the server increases your rupee count", which is the anti-dupe change.
- **1.38 changes how weapons are registered.** Up to 1.37 the client uploads each weapon it holds with id 25 (name, image and the full script, in the same layout as `PLO_NPCWEAPONADD`). From 1.38 it sends id 33 as `chr(0)` + `chr(default weapon id)` instead, and the server answers default weapons with 43 `DEFAULTWEAPON`. Both confirmed live (25 in 1.33, 33 in 1.39).
- **1.39 adds the profile window**, with request (80), set (81) and reply (75).

So a server can only tell 1.38 and 1.39 apart from the earlier builds by these ids or by the token. gs2lib also knows `GNW02070` (1.38 rev1) and `GNW12080` (1.39 rev1). Neither build is in the archive.

What the player sees change

HUD art (`state.png`, `sprites.png`)

- **1.33 and 1.34.** `state.png` is 224×81, and the three item buttons carry the letters **A**, **S** and **D**. The HUD digits 0–9 sit at the bottom of `sprites.png` (128×484).
- **1.35.** A small **CHAT** label is added to `state.png`. It marks the focused player in 2/3/4-player mode, a feature the 1.34 notes describe.
- **1.38.** The button letters are removed from the art, because keys became configurable in that release. The client still draws A/S/D on the buttons itself (seen live in 1.39). The digits move from `sprites.png` (now 128×452) into `state.png` (now 224×97).

A client that copies these looks needs the A/S/D button art up to 1.37, and blank button art with the letters drawn on top from 1.38. It also has to read the digits from the right sheet.

Tiles (`pics1.png`, always 1856×512)

The tileset is unchanged from 1.33 to 1.37. 1.38 changes 81 tiles ("some new tiles in pics1.png") and 1.39 changes 7 ("added some tiles and corrected the tile type for the dirty water").

Windows and options

Build	Change
1.33	Player list (F7) with four sections (GPs, Guild, Normal, Ignore). Its popup menu has Message, Ignore and History. More Options has 13 checkboxes: vsync, joystick, music, sound, reverse stereo, save downloaded images, external chat, compatibility draw, hide decompression errors, fixed UDP port, no UDP, ignore mass messages, no DirectInput.
1.34	The F6 debugger is rebuilt with Next, Run, Stop next call and Stop next script, plus panes for this.vars, vars, flags and NPC attributes. It only works offline.
1.35	Player list popup gains **Buddy** (`buddyaccounts.txt`). More Options gains "Don't use HTML for PMs". PMs render as HTML. The editor gains "Generate map" *[inf]*. A "(Senior GP)" tag string appears.
1.36	More Options gains "Limit the length of nicknames to [50]". An "(RC)" tag string appears.
1.38	The editor gains a "Show blocking tiles" checkbox. "Support for external chat programs" and its in/out message files are removed.
1.39	Player list gains **Profile** and **Options** buttons and a Profile popup item. There is a new Player List Options window (ignore mass messages, no HTML, no images in PMs, sort by nickname). There is a new Profile window (real name, age, sex, country, ICQ, e-mail, homepage, favourite hangout, favourite quote). More Options swaps the moved PM options for "Don't connect levels", "Ignore toalls" and "Don't load heads for the playerlist / map".

Controls

- **Up to 1.37** the keys are fixed: arrows or IJKL, D item, S or Space attack, A action, M map, Q item list, P pause, Tab chat.
- **From 1.38** each action can have up to three keys, set in `graalconfig.txt`. The actions are `up`, `down`, `left`, `right`, `item`, `attack`, `action`, `map`, `switchtochat`, `itemlist` and `pause`. The usable key names

are A–Z, 0–9, Tab, Enter, Shift, Space, P_Up, P_Down, the arrows, N0–N9, Add, Sub, Mul, Div, Dec, and F9–F15.
- **1.38** adds Alt+8 and Alt+9 for zoom, and Ctrl+letter emotion icons from `levels/emotions` for A, B, D, E, F, G, H, L, M, O, P, R, S, T, W, X and Z. AFK, BRB and Zzz stay until the player moves.
- **1.39** adds Ctrl+C, K and Y. It also adds chat-line history on Up and Down (100 lines), and lets the player click the weapon list to select.

Gameplay and rules (release notes)

- **1.34.** Hitting a mounted player costs their horse one bush, and at zero bushes the rider is pushed off. Two players can share a horse offline. Sounds are limited to one per type per frame and eight at once. `canbepulled` is separate from `canbepushed`.
- **1.36.** The server cap rises from 220 to 28000 players. Scripts can read the level object arrays (`bombs[]`, `arrows[]`, `items[]`, `explos[]`, `horses[]`).
- **1.37.** Sound volume falls off with distance. `showimg` works on normal NPCs. `playlooped` and `stopsound` are added. Arrays can be initialised as literals, and `true`, `false` and `hasweapon()` are added.
- **1.38.** A nickname equal to the account name shows as `*nick`. Item pickup is checked by the server. A laid-down shield is visible to others. Scripts can read and write `npcs[]`, and `hitnpc`, `setlevel2`, `changeimgpart` and the `test*()` functions are added. Explosions are capped at 1024. The `gotbow`/`gotsword` flags are dropped. PMs are saved to `pms/`.
- **1.39.** Head images are handled differently. Dead players don't block. Weapon-list and M-map fixes. In the editor, tiles can be painted by dragging, and an image dropped on the level becomes an NPC.

Script commands

The GS1 command table grows from 88 to 93, 93, 104, 106, 109 and 111 commands. The full lists are in the TOML, and the per-build additions are in `cross-version-timeline.md`. Two of the 1.34 additions, `carryobject` and `throwcarry`, are only announced in the 1.37 notes. So being in the table does not prove a command works in that build.

Other shipped files

- **Languages.** English, Deutsch and Portugues up to 1.36. Español and Italiano are added in 1.37, Swedish in 1.38 and Français in 1.39.
- **World maps.** The world map image grows from 616×784 (1.33) to 952×952 (1.36) and 1008×952 (1.37). `mappw` is added in 1.35 and `mapmantara` in 1.37. These are server content, not client features.
- **gif2mng** ships from 1.38.

Using the table

- `[[version]]` holds the full values for one build. `[version.features]` holds that build's on/off flags, and `[version.assets]` describes which HUD and tileset revision it ships.
- The top-level `[features]` table gives each flag's first build, its confidence and a one-line note. `[confidence]` gives one line per key on how sure each value is.
- To pick ids, use `pli_sent` for what the client may send, and `plo_known`, `plprops_known` and `npcprops_known` for what it parses. A server that sends anything else to one of these builds hits the default branch, and the client ignores it. Confirmed live in 1.33 and 1.39 for ids 47, 82, 156, 168, 190 and 194.

Live check (2026-10-30)

1.33 and 1.39 were run under Wine against a local GServer-v2 beta4 with a local listserver copy, on loopback only, and the traffic was decoded.

- **Login props.** After the level loads, the first id 2 carries props 0 (nick), 11 (head), 13 (colours), 18 (status), 31 (UDP port) and 33 (additional flags), the same in both builds. The head goes as `chr(head number)` when it is a numbered default head. 1.34 to 1.38 are assumed to match.
- **Steady traffic.** Id 31 `PACKETCOUNT` goes about every 10 s. Id 3 NPC props go for each level NPC after the load, and level scripts cause id 18 and 7.
- **Board and files.** As described under "What is the same in all seven". For a name sent without an extension the server adds `.gif`. Downloaded images were not written to disk in this setup, and the level was cached under `weblevels/<server>/`.
- **Server message.** 1.33 shows the server message (41) as raw HTML text.

Server compatibility (GServer-v2 beta4)

- It needs `generation = classic`, with the GNW tokens listed in `allowedversions.txt`. Otherwise it refuses the client with "Allowed: 2.0 - 2.31".
- The server checks the account with a listserver, even though the client connects directly. Without one the client is told "The login server is offline".
- It kicks 1.33 for id 25 ("Unknown Player Packet: 25"). 1.34 to 1.37 send the same id (*inferred* to be kicked too). A proxy that drops id 25 lets 1.33 play, though beta4 then warns about an invalid packet count, because the dropped messages are missing from the count.
- 1.39 logs in and plays. 1.38 has the same send set apart from profiles (*inferred* to work).
- It sends prop 10 as a gani string, which these clients misread (see "Prop encodings").

Open

- Login props in 1.34 to 1.38 (seen live only in 1.33 and 1.39).
- Per-prop byte layouts for props not seen live (assumed above).
- The client → server UDP id set. The receive set is confirmed.
- The exact letter-to-icon mapping for emotions. The names are built at run time, and only AFK (A), BRB (B) and Zzz (Z) are given in the notes.

`client-internals/water-and-ganis.md`

Water, splashes and the built-in animation names

Date: 2026-10-28. Read from the 2.05 rev 2 client and checked against its executable's string table. **[2.05]** These notes contain no addresses, routine names or decompiler text.

Built-in animations

The client keeps one table of animation names, which it plays as `<name>.gani`. In table order:

#	name	#	name	#	name
0	idle	11	carrystill	22	hurt
1	walk	12	hiddenstill	23	ridehurt
2	walkslow	13	carry	24	dead

#	name	#	name	#	name
3	kick	14	hidden	25	maps1
4	sword	15	shoot	26	maps2
5	ridesword	16	ridestill	27	maps3
6	spin	17	ridefire	28	swim
7	push	18	rideeat	29	lava
8	grab	19	ride	30	carried
9	pull	20	sit		
10	lift	21	sleep		

Uses that were seen in the code:

- The sword swing plays `sword`; on a horse, `ridesword`.
- The spin (a second S press during a swing, with the has-spin status) plays `spin`.
- The A key's grab plays `grab`, and its lift plays `lift`.
- The item key's "lay item" mode plays `hiddenstill`.
- Firing a weapon on a horse plays `ridefire`.
- A throw sets the player back to `idle`.
- Swimming plays `swim`, or `lava` on a lava tile (type 12).

Water

The client checks the player's tile on every game tick. It skips the check while the sword swings, while riding, and when the player is not under local control.

- **Into a liquid tile.**
  - When not swimming already, the client starts a splash, which is break effect 5, at the player's position.
  - When the player was not walking at the time (a warp or a knock), it starts two more splashes.
  - It then enters swim mode with `swim`, or `lava` when the tile type is 12.
- **Off a liquid tile while swimming.** One splash, then back to standing.
- **Liquid test.** The lava animation is chosen inside the water branch, so the client's liquid test is true for lava tiles too. **[inf]** The test itself is a jump table that was not followed.

Break effect sounds

Each break effect plays a sound as it starts, at `(x + 1, y + 1)` from its anchor:

- `sounds/water.wav` for the splash (effect 5);
- `sounds/crush.wav` for every other effect.

`client-internals/zelda-online-1998.md`

Zelda Online (1997–98) — precursor analysis

Date: 2026-10-14 Author: Stefan Knorr (knorr@informatik.uni-rostock.de) Copyright: GServ Standalone (C) Stefan Knorr 1997; Zelda Client for GServer (C) Stefan Knorr 1998

What this is

The Java Zelda Online local version that predates the Delphi Graal client. The server is a text-line lobby; the client is a full Zelda-style game that loads `.zelda` levels and optionally joins multiplayer on port 14801.

Layout (promoted to shanalysis/zelda-online-1998)

Path	Contents
`server/`	`GServer.java`, `Player.java` — lobby server
`game/`	`Zelda.java`, `ZeldaMain.java`, `ZeldaClient.java`, `MapPanel.java`, `Computer.java`, `FileLink.java`
`programming/`	`makeArrays.java`, `addSprites.java` — generate `arrays.dat`
`readme.txt`, `readme-2.txt`	original notes

Levels (51 `.zelda`) and binaries (class, gif, arrays.dat, audio, bat) moved to `/recycler/trash/`.

Server protocol (text lines, port 14801)

Prefix	Meaning
`N` / `O` / `S`	game#name (normal / observer / super)
`I` + 8-char id	server assigns id
`+` / `-`	enter / leave available list
`P` + id	play against
`C` + id	disconnect from party
`D` + data	game data relay
`V` + id	observe

Player ids are 8-char (`0000` + 4-digit). Game data is relayed between opponents; no binary framing.

Level format (`.zelda`)

Header: `Z3-V1.02` or `Z3-V1.03` (8 bytes).

Board: 64×64 tiles, RLE. Code bits = 11 (v1.02) or 12 (v1.03). High bit set means count; otherwise tile id. Double-repeat flag for alternating pairs.

After board: - text lines of `FileLink` records (`filename x y w h newx newy`) until `#` - binary computer records (4 bytes each: x, y, type, stop) - text sign records (`signx-32`, `signy-32`, coded text)

Tile / sprite tables

`makeArrays.java` writes `arrays.dat` with: - `nowall` (passable tile ids) - `swamp`, `jumpstone`, `bushobj` / replace, `liftobj` / replace, `chestfields` / replace, `movestone` - sprite placement tables (`spritevalues`, `hurtsprites`, `compusprites`, etc.)

`addSprites.java` appends the sprite crop table (x, y, w, h packed into ints).

Client connection

`ZeldaClient` connects to host:14801. On enter it sends `N` + level-prefix + `#def`. Subsequent `D` packets carry position, chat, nickname, sprite colors, computer moves, board modifies, extras, carries, shots. Leader election is by earliest login time.

Relation to later Graal

Zelda Online	Later Graal
port 14801	listserver / world ports evolve
text-line lobby	binary framed protocol with version token
`.zelda` RLE board	`.nw` GLEVNW01 text board
`arrays.dat` tile tables	`tiledefs.txt` / pics1.png
Java applet	Delphi client

This is the last point before the version-token and binary-packet era.

See also

`precursors-and-fan-builds.md` (2026-10-29) covers both Java versions from the uploads in more depth. Zelda Online 1 is a single-player applet with no network code. Zelda Online 2 ships 51 levels. It also covers the 11-bit `Z3-V1.02` code width, the full GServ line set, password saves, the ZeldaEditor and how its dialogs became 1st Graal's editor, and Java Graal.

`version-profiles/2.22-2.31.md`

Version profiles: 2.22, 2.3 and 2.31

Date: 2026-10-22

This note covers what a client has to do to *be* 2.22, 2.3 or 2.31: what it sends and understands on the wire, and how it looks and behaves for the player. The machine-readable table is `2.22-2.31.toml`.

Sources: - `Graal_222.exe` and `Graal_231.exe` from the Graal Reborn 2.2.2.0 folder, read statically. That covers form resources, embedded images, strings, the GS1 symbol table, and the send and receive paths. - The seihai-lab captures (2.22 only). - GServer-v2 source. - The release notes shipped in the same folder.

No 2.3 executable was uploaded, so every 2.3 value is inferred.

Tags: **[bin]** client binary · **[cap]** lab capture · **[srv]** GServer-v2 · **[doc]** release notes · **[inf]** inferred.

Builds in this range

Label	FileVersion	Token	Binary	GServer-v2 default `allowedversions.txt`
2.22	2.2.2.0	`GNW03014`	yes	enabled
2.3	2.3.0.0	`GNW14015`	**none**	commented out
2.31	2.3.1.0	`GNW28015`	yes	commented out

A lab server needs `GNW14015` and `GNW28015` uncommented before it will let 2.3 or 2.31 in. **[srv]**

Caveat on the binaries: the Reborn folder's `Graal_222.exe` has a 2009 file date and `Graal_231.exe` a 2007 one. The 2.17 build in the same folder is rebranded "Eradia Online", so a community edit of the other two cannot be ruled out. Any edits would most likely be to text and server-list strings, not to the protocol code. **[bin][inf]**

Wire protocol

The three builds share the gen-5 profile already written up in `protocol/client-versions.md` and `protocol/codec-and-framing.md`: - Login type 5, a key byte, the token, the account and the password. - The login frame is zlib with no compression byte. Every later frame carries a compression byte: 2 below 0x30 bytes (4 if zlib is shorter), 4 below 0x1000, and 6 (bz2) from there. - XOR limits are 12 dwords for type 2 and 4 for types 4 and 6. These are caps on the group count. The count itself is one LCG step per 4-byte group present, so a body of 5 bytes gets 2 steps, and a partial last group still steps (`protocol/codec-and-framing.md`). - Listserver id 7 carries `GNW30123newmain`, with hosts from `arrays2.dat`.

2.31's send path is the same code as 2.22's: same size thresholds, XOR limits, and 0x40000 queue cap. **[bin]**

	2.22	**2.3**	**2.31**
Server → client ids handled	80	as 2.31 [inf]	81: drops 44 `HASNPCSERVER`, adds 84 `LARGEFILESIZE` and 189 `MOVE2` **[bin]**
Client → server ids sent	0–24, 26–30, 32–35, 37–41, 43, 44, 80	as 2.31 [inf]	2.22 minus 21 `PUTNPC`, plus 46 `ENTERLEVEL` (sent on every level entry including doors, links and server warps, lab **[cap]**; 2.22 sends no 46) **[bin]**
Highest PLPROP / NPCPROP parsed	74 / 73	as 2.31 [inf]	80 / 77 (X2, Y2, Z2) **[bin]**
Precise positions	no	yes **[srv][doc]**	yes **[srv][bin]**
After login	37, then 39 (10 bytes)	same	same **[bin]**

Notes on the table: - **Client → server set.** It was read from the call sites of the client's send helper. Id 44 (`PROCESSLIST`) was also seen in the 2.22 capture, sent as the reply to server id 182. Ids 36, 42 and 45 are not matched to a call site, so treat the set as a floor, not a complete list. **[bin][cap]** - **Tampercheck (39).** 10 raw bytes, identical at every login of a build and across accounts and login keys, so a per-build constant. **[cap]** 2.22 `250e4d0f962b03ede631` and 2.31 `291655b7602e87c9eb82` (2.17, same layout: `2a7bad195e231243d82c`). GServer-v2 reads and discards it, so the lab cannot say whether a different value would be rejected. The bytes are two chained CRC-32 words over the client's own memory image in a 5-byte non-GINT5 form (`client-internals/decomp-findings-2026-10-24-b/01-tamper-check-id39.md`; words in `tampercheck-words.tsv`). The words were reproduced offline for 2.17.1, 2.22 and 2.31. Re-sends differ by build: 2.17 once at login; 2.22 at login, on every gmap edge crossing and after every client-initiated level-link request (not after a server-side warp), with no periodic send; **2.31 at login and then about every 3.2 s**. The 3.2 s cadence is a 50 ms timer counting modulo 64 (`client-internals/decomp-findings-2026-10-24-b/03-periodic-ids-2.31.md`). 2.31 additionally sends `PLI_SHOWIMGPLAYER` (24, a 1-byte payload) about every 5.1 s and `PLI_ENTERLEVEL` (46, 1 byte, value 1) on every level entry (lab, batch b2 and c; not once at login). 2.22 also sends `PLI_SHOWIMGPLAYER` (24) = 9 about every 5.1 s, but only while another player is in its level. The meaning of the id 24 payload is unknown. 2.31 also answers the server's process-list request with `PLI_PROCESSLIST` (44), as 2.22 does. **[cap]** - **2.22 login sequence in the lab.** The client sent 37 `English`, then 39, then 34 `UPDATEFILE` for its default images (sword, shield, body, bomb), 23 `WANTFILE`, and 2 `PLAYERPROPS`. It answered the server's 182 with 44. The `gr.x` / `gr.y` flag sets in the capture come from the server's movement NPC, not from the client. **[cap]** - **Zlib fix NPC.** GServer-v2 sends a `-gr_zlib_fix` weapon to every client from 2.21 to 2.31. It pads chat text made of long runs of one or two repeated characters. This is a server-side workaround for a client

compression bug, so a faithful 2.22–2.31 client should keep that bug or at least tolerate the padding. **[srv][inf] - 2.31 and `setmap`.** 2.31 does not request the map file after `setmap`, so GServer-v2 pushes every BIGMAP file at login for 2.31. **[srv]** It does not do this for 2.3, which suggests 2.3 still asked for the file. **[inf] - Precise movement.** From 2.3, GServer-v2 puts X2/Y2 before X/Y in other-player props and stops its `gr.x` / `gr.y` / `gr.z` movement-flag workaround. **[srv] - UDP.** Present in both builds, with the `noudp` and `fixedudpport` keys and the "Don't use UDP for data packages" option. `noudp` defaults to on. **[bin]**

What the player sees

Start window and server list

	2.22	**2.31**
Login and server-list backdrops	the "GraalOnline" banners (samurai house, then the character line-up)	**identical images [bin]**
Start-window buttons (seen live)	Training · Options · Get an account! · Start	Training · **Editor** · Options · Get an account! · Start. Editor launches `GraalEditor.exe`. **Training is still there in 2.31 [cap][doc]**
Server-list tabs	Gold · Silver · Bronze **[bin]**, possibly a Reborn edit	Graal Gold · **Graal 3D** · Classic · Hosted. A 3D server shows "You need to download Graal3D…" or starts `Graal3.exe` **[bin][doc]**
Tab for the same lab server (live)	**Bronze**	**Hosted**. Only tabs that hold a server are drawn. **[cap]**
Server-list buttons (live)	Show More · Donations · Connect	same **[cap]**
`graal://` and `graal3://` links	no	registered at start-up. A link sends `serverstartparams` to the server's Control-NPC **[bin][doc]**

For comparison, 2.17's tabs are Graal Gold · Classic · Hosted, and its start buttons are the same as 2.22's. **[bin]**

In-game windows and keys

- **The editor is gone from the 2.31 client.** 2.22 still carries the full level editor:
  - the toolbar (New, Open, Save, Cut, Copy, Paste, warp field, links, signs, flags, Export Tiles Array, Play)
  - the tile and object panels and the "Show blocking tiles / foreground / terrain grid / Hide npcs" checkboxes
  - the dialogs for chests, baddies, NPCs, signs, links, flags, characters, object names and the tile array
  - the F6 script debugger

  2.31 has none of these forms (18 forms instead of 29). **[bin]**
- **The F1 window, seen live.** 2.22 shows six buttons: General options, Edit level (F4), Load level (F5), Debugger (F6), Playerlist (F7) and Serverlist (F8). 2.31 shows four: General options, Load level (F5), Playerlist (F7) and Serverlist (F8). Edit level and Debugger are still in the form resource but hidden at run time, and pressing F4 in game does nothing. **[cap][bin]**
- **Options → General**: 2.22 has "Save downloaded images (to)". 2.31 replaces that checkbox with "Use filename cache (don't scan folders each startup)", config key `cachefilenames`, backed by `FILENAMECACHE.txt`. Every other tab is unchanged: Video, Audio (midi / mod-mp3 / radio / voice / sfx), Microphone (GSM / LPC / LPC-10 voice chat), Controls (joystick, DirectInput, Change keys), and the player-list options. **[bin]**
- **Key layout**: the Change Controls form is identical in the two builds. 2.31 fixes the saving and loading of the key configuration. **[bin][doc]**

- **HUD**: identical in a live session on the same server: A / S / D item slots, the rupee, bomb and arrow counters, the spinning heart icons and the MP bar. **[cap]** There is also no difference in the shipped images or the HUD image file names (`state.png`, `letters.png`, `sprites.png`, and so on). The two status icons `ss_arrow.png` and `ss_bomb.png` are no longer referenced in 2.31, and neither are `graalmap.png`, `headsconfig.txt`, `loadgmaps.txt`, `recordpos.txt` or `shield1.gif`. All of these belong to offline or editor play. **[bin][inf]**

Game behaviour

	2.22	2.3 / 2.31
Classic client-side mode (client applies NPC props itself when the server has no NPC server)	yes, honours `HASNPCSERVER`	removed **[bin][doc]**
Smooth positions	half-tile X/Y	full-precision X2/Y2/Z2 for players and NPCs **[bin][srv]**
NPCs crossing gmap level borders	buggy	fixed in 2.31 **[doc]**
Float-to-string in scripts	buggy	fixed in 2.31 (`#v` prints integers with more than 6 digits) **[doc]**
CPU use	about 99 % (busy timer)	new timer **[doc]**
Object lookup	linear	B-tree. Huge ganis are no longer culled early **[doc]**
Local editor flags (`local.`, `this.-`, `gotheart_`)	present	gone with the editor **[bin]**

GS1 script differences (from the client's symbol table) [bin]

Change	**2.17 → 2.22**	**2.22 → 2.31**
Added	`gravity`; `textwidth(zoom,font,style,text)`; `textheight(zoom,font,style)`; `isthisflagset` and `isplayerflagset` (replacing `isflagset`); one new # message code with a string argument (`#R`)	`timevar2`; `downloadsize`; `onwall2(x,y,w,h)`; `onwater2(x,y,w,h)`
Removed	`isflagset`	none (the command list is unchanged)

`downloadsize` is announced in the 2.22 notes, but it is only in 2.31's symbol table, which fits the 2.3 note "is correctly giving the size". A 2.22 profile should treat it as undefined (0). **[bin][doc][inf]**

Using this in Seihai

The `features` block in the TOML gives one switch per difference above. The three that change the most for the player:

1. `editor_in_client` (2.22 only): F4 and F6, and the Edit level and Debugger buttons in F1. Training and Load level (F5) stay in both.

2. `serverlist_tabs`, `graal3d_tab` and `url_protocols` (start window).
3. `precise_move` (2.3+), with the matching PLPROP 78–80 and NPCPROP 77 parsing.

Screenshots from the 2026-10-30 live run (beta4 lab server) are in the project's `graal-decompile/live-trace/shots/v222-231/`.

Open

- A 2.3 binary, to confirm that 2.3 matches 2.31 apart from its bug fixes.
- Call sites for client → server ids 36, 42 and 45 (the 2026-10-24 decompile pass covered the receive handlers and the tamper builder, not these send sites).
- Whether the 2.22 tab names Gold, Silver and Bronze are original or a Graal Reborn edit.
- The meaning of the 2.22 `#R(string)` message code.

`version-profiles/3.04-3.153.md`

Version profiles: 3.04 (Linux) and 3.153 (Mac OS X)

Date: 2026-10-30

This note is the per-version profile for the two readable Graal 3.x clients, in the same shape as `2.22-2.31.md`. The machine-readable table is `3.04-3.153.toml`. Background on the clients themselves (Torque engine, licence file, listserver, file formats) is in `protocol/v3x-clients.md`.

Sources: the 3.04 Linux client (unpacked from UPX) and the 3.153 Mac OS X client, read statically, plus the compiled TorqueScript and prefs shipped with them. 3.04 was then run against GServer-v2 in the lab (`lab-304-linux-2026-10-30.md`). Tags: **[bin]** client binary · **[file]** shipped data · **[cap]** lab run (3.04 only) · **[inf]** inferred.

Builds

Label	Platform	Token	Windows twin
3.04 (rev 1, 3.041)	Linux i386	`G3D03014`	2.22 (`GNW03014`)
3.153	Mac OS X PPC	`G3D28015`	2.31 (`GNW28015`)

The tokens share date stamps with the Windows builds, and the wire protocol matches: gen-5 login (type 5, key byte, token, plain account and password), the same LCG cipher, framing, compression thresholds and XOR caps. **[bin]**

Login and listserver

- Listserver: id 7 `G3D30123newmain` and id 1 (account, password), sent together in one plain zlib frame; the reply is gen 5 **[cap]**. The host list comes from the signed `license/license.graal`, which names `<list-host-3x>:14922`. For Seihai the profile uses `<list-host-reborn>:14922` (supported) with port status **assumed**: no G3D token has been tried against the public listserver. **[bin][file]**
- The game login frame also carries 37 (language) and 34 for `bigmap.txt` and `minimap.txt`. The client **never sends 39** (tampercheck) and sends nothing periodic over TCP; its 5.1 s message (2.31's TCP 24) goes to other players over UDP. Then it starts UDP unless `$pref::Graal::noudp` is set, and requests its own images with 23 and 34, at most five at a time. **[bin][cap]**
- First props sent: 0, 18, 31, 33, 52, 75 (`linux`) and 76 (1250 in the lab). **[cap]**
- Server names carry a category prefix: `P` Gold, `3` 3D, `H` Hosted, `U` hidden, none Classic. **[bin]**

Client → server ids

	3.04	3.153
Sent	0–24, 26–30, 32–38, 40, 41, 43, 80, 81	the 3.04 set plus 46, 50, 93, 96–98, 152, 154–158
Complete?	yes, every game-code send site	no: 2, 3, 15, 24, 29 and 80 are assumed from 3.04 **[inf]**
Versus Windows twin	no 39 or 44; adds 36 and 81	no 39 or 44; keeps 21 (2.31 dropped it); adds 36, 50, 81 and the RC/file ids

New in 3.153: 46 ENTERLEVEL, 50 RAWDATA, 93/96/97/98 the RC file browser (up, move, delete, rename), 152 REQUESTTEXT, 154 SENDTEXT, 155/156 large-file upload start and end, 157 UPDATEGANI, 158 UPDATESCRIPT. **[bin]**

Server → client ids

3.04 handles 79 ids, 3.153 handles 88; the table is in `protocol/v3x-clients.md`. Against 2.22, 3.04 lacks 170, 177, 182 and 183 and adds 84, 101 and 102. **[bin]**

Props

	3.04	3.153
Own props written	0–76 except 25, 27–30, 34, 36, 50, 51	0–80 except 25, 27–30, 34, 36, 50, 51, 77
Own props parsed (PLO 9)	0–74 except 33, 50, 51, 53	0–80 except 33, 50, 51, 53, 75–77
Other players parsed	0–76	0–80
NPC props parsed	0–73 except 48–52	0–77 except 48–52
Precise move (X2/Y2/Z2, MOVE2)	no	yes

Prop 75 is OSTYPE and carries `linux` or `mac`; prop 76 is TEXTCODEPAGE. The written-prop exclusions are the same as 2.17's. **[bin]**

Player-visible features

- **3.04** start window: Nick, Language, Options, Tutorial, Account, Password, Get Account, Start. Server list tabs Graal Gold, Classic, Graal 3D and Hosted, with a direct-connect field. Options tabs Graphics, Audio, General. Player list, PMs, a profile window, and an offline mode that loads a local level. No in-client editor. **[file]**
- **GS1** (3.04): 496 registered names. Every 2.05 command is present except `putobject`, `setbackpal`, `setbody`, `setbacktile`, `drawovertrees` and `setbacktile2`. 3.153 was not counted. **[bin]**
- **3.153** adds a 3D mode (Torque `GameConnection` on UDP listed port + 1, `Graal3D` protocol 19, with missions and terrain; see `torque/wire-by-build.md`), snow and xmas themes, the RC file browser and script uploads, and server-side class scripts (131–134). The shipped prefs turn UDP off (`noudp = 1`, fixed port 14899). **[file][bin]**

Observed behaviour (3.04)

Walk speed 10 tiles per second, `push` after about 0.9 s against a wall, a 0.25 s `sword` gani, dropped items removed by the client after 8.3 s, no 11 when a bush is thrown, chat drawn as white text with a blue shadow and no bubble, and the 2.05 UDP channel unchanged (movement, chat, trigger action, a 5.1 s keepalive). Details and timings are in `lab-304-linux-2026-10-30.md`. **[cap]**

Open

- None for respawn offline: 3.04's offline mode cannot cut, lift or break classic objects at all (lab note).
- Try the `G3D30123` listserver login against the public listserver to settle the port status. This is an outward action and has not been done.
- A clean copy of the 3.151 Linux installer would give a third data point between the two.

`version-profiles/lab-304-linux-2026-10-30.md`

Lab run: the real 3.04 Linux client on GServer-v2

Date: 2026-10-30

The 3.04 Linux client (the original UPX-packed executable from the installer) was run in a cloud container: i386 libraries, Xvfb with Mesa software GL, and a network namespace with loopback only. `/etc/hosts` in the namespace pointed `<list-host-3x>` at the loopback address, so the signed `license.graal` did not need editing. The servers were GServer-v2 (with `G3D03014` added to `allowedversions.txt`) and the lab listserver, extended to accept the 3.x listserver login. One online session ran for about 5 minutes. A short offline session followed. Tags as elsewhere: **[cap]** seen on the wire or on screen in this run.

Setup, for repeating it

- Packages (Ubuntu 24.04, i386 multiarch): `libc6`, `libx11-6`, `libxext6`, `libgl1`, `libglx-mesa0`, `libgl1-mesa-dri`, `libglu1-mesa`, `libopenal1`, `libasound2t64`, plus `xvfb`, `xdotool` and `iproute2`. The client ships its own SDL 1.2 and fmod in `lib/`, and `libstdc++` is linked in.
- Run it from the install folder with `LD_LIBRARY_PATH=./lib`, inside the namespace, on an Xvfb display of 1024×768×24 with GLX. It opens at 800×600 and keeps its prefs and `console.log` under `~/.garagegames/graalexe/`.
- In the namespace, bring `lo` up before starting the servers. GServer-v2 needs `G3D03014` in `config/allowedversions.txt`. The lab listserver must accept id 7 in the first plain frame and answer in gen 5.
- The client scans its folders only at startup, so a level copied in later is not found until a restart.
- The UPX stub leaves the running process named `3`, so `pkill graalexe` matches nothing. Stop clients by pid.
- A second client needs its own `HOME` (prefs live under `~/.garagegames`) and its own X display.

Startup and listserver

- The client runs as shipped. Sound fails to open (no device) and the game carries on. The start window, server list and in-game screens match what the TorqueScript files describe. **[cap]**
- **Listserver login is one frame.** The client sends a single plain zlib frame holding both id 7 (`key`, `G3D30123newmain`) and id 1 (account, password). Static reading suggested a flush between them; the wire shows none. The lab listserver answered in gen 5 (compression byte plus partial XOR, keyed with that key), and the client showed the list and the status line. **[cap]**
- The server list shows only the tabs that have servers. A server name with no prefix appears under Classic. **[cap]**

Game login

- The first frame is plain zlib (no compression byte, no XOR) and holds four messages: 5 login (type 5, key, `G3D03014`, account, password), 37 `English`, then 34 UPDATEFILE for `bigmap.txt` and `minimap.txt`. GServer-v2 logged it as "New Client (2.22+)", version 3.041. **[cap]**
- Every later frame is gen 5. The server's first reply used bz2 (compression byte 6) and the client decoded it. **[cap]**
- First props message (id 2) after the level arrives: 0 NICKNAME, 18 STATUS, 31 UDPPORT (a random port, as `fixedport` is 0), 33 ADDITFLAGS 0, 52 PLANGUAGE `English`, 75 OSTYPE `linux`, 76 TEXTCODEPAGE 1250. **[cap]**
- Then 34 UPDATEFILE for the player's head and the two default weapon icons, and 23 WANTFILE for the player-list status icons named by server id 180. These come in two batches, which matches the five-request queue limit. **[cap]**
- **No 39 (tampercheck) in the whole session, and no periodic message over TCP.** In five minutes of mostly idle play the client sent only 1, 2, 13, 18 (GServer's movement script), 23 and 34. 2.31 sends 39 about every 3.2 s and 24 about every 5.1 s. The 5.1 s message does exist in 3.04, but it goes over UDP to other players (see the two-player run below), and this session had no other player. **[cap]**

Movement and animation

- Walking moves 0.5 tiles per 50 ms (10 tiles per second). While moving, the client sends X or Y every 100 ms. When the key is released it sends `idle` about 0.1 to 0.15 s later. **[cap]**
- **Pushing against a wall:** the `push` gani starts about 0.9 s after the player stops against a blocking tile. Releasing the key after 0.6 s did not trigger it. **[cap]**
- **Chat:** the text is drawn above the head in white with a one-pixel blue shadow down and to the right. There is no bubble or box. Tab opens the chat bar along the bottom of the screen. **[cap]**
- Key bindings (Torque `config.cs`): arrows move, A, S and D are the item keys, Q, P and M open windows, T is the microphone, F3 options, F5 load level, F7 player list, F8 server list. **[file]**

Bushes, lifting, items

- **Sword on a bush:** the client sends 34 UPDATEFILE for `sword.gani`, then 1 BOARDMODIFY (2×2 tiles), then gani `sword`. The `sword` gani lasts 0.25 s before `idle`. The server echoes 7 and adds a dropped item with 22. **[cap]**
- **The dropped item vanished after 8.3 s.** The client sent 13 ITEMDEL for it on its own, without being picked up. **[cap]**
- **Respawn online is the server's job.** GServer-v2 restored the bush with 7 BOARDMODIFY 14.6 s after the cut (its `respawntime` is 15). The client only draws what the server sends, so this lab cannot give the official server's respawn time. **[cap]**
- **Lifting a bush:** 1 BOARDMODIFY at once, gani `lift`, carry sprite 1 after 0.2 s, `carrystill` after 0.3 s, and `carry` while walking. **Throwing** sends only `idle` with carry sprite −1. No 11 THROWCARRIED was sent for a thrown bush. **[cap]**

Offline

- Offline mode (the Tutorial button) loads `levels/tiledefs/tiledefsOffline.txt`, one line: `picso.png,offlinetutorial,1,0,0`. `picso.png` is a 1024×512 sheet of terrain only (water, sand, grass, rock, snow, ice). It has no bushes, pots or liftable stones, so the tutorial levels have none either. **[file]**

- A test level (`labbush.nw`: grass, with four each of classic bush 2/3/18/19, stone 34/35/50/51 and vase 684/685/700/701 tiles) was loaded offline with F5. The objects draw wrongly (each 16-pixel cell shows a whole small object, and the bushes and stones alternate by position whatever their tile ids), the player walks through them, and the sword does not cut them. The same happened with the offline tiledefs file removed. Loading the GServer world level offline gave the same kind of broken picture. **[cap]**
- So **3.04 has no offline respawn to measure**: offline, the classic cut, lift and break objects do not work at all. Online, the server does the respawn (above). **[cap]**

Two players: the UDP channel

Two clients (separate prefs folders, one Xvfb display each) logged in to the same level. Both had `noudp = 0`, as shipped for Linux. The decoder is the 2.05 one from `protocol/udp-p2p.md`, unchanged. **[cap]**

- **Same channel as 2.05.** Each datagram is `seq` plus `GCHAR id` plus fields, XORed whole with `recipientId XOR 1`: key 3 to player 2, key 2 to player 3. `seq` counts from 0 per sender and recipient pair. There is no zlib and no length prefix.
- **Id 0, movement:** X×2, Y×2, SPRITE, STATUS (21), CARRYSPRITE (−1), and the first letter of the current gani (`W` walk, `I` idle, `S` sword). It is sent every 100 ms while moving, then once on stopping. The same position also goes to the server over TCP as props.
- **Id 15 with one byte, 9: sent about every 5.1 s to each peer.** This is the message 2.31 sends over TCP as 24 every 5.1 s with the value 9. In 3.04 it goes to peers over UDP and not to the server.
- **Id 22, chat:** the raw text. The same text also went to the server as prop 12 over TCP.
- **Id 26, trigger action:** a mouse click sent `0 0 0` (an NPC id of 0), X×2, Y×2 and the action `leftmouse`. It went over UDP only, with no TCP copy.
- In both two-player runs, one of the two clients stopped sending UDP to its peer after 11 to 20 s, and its later chat went over TCP only. The trigger for this was not traced.
- The other player's name shows under them in blue. While a client window has no focus the name reads `name (paused)`.

Not yet tested

- Pots and stones online (they need a level with them in reach and enough glove power), PMs, and the profile window.
- Why one client stops sending UDP after a few seconds.

`version-profiles/lab-mac-tiger-2026-10-31.md`

Lab run: the real Mac 4.20r6 and 3.153 clients on Mac OS X 10.4 in qemu

Date: 2026-10-31. Tags as elsewhere: **[cap]** seen on the wire or on screen in this run. Raw captures (proxy logs, pcaps, screenshots, the qemu driving scripts and `aglshim.c`) are in the project files under `v4/lab-mac420r6-2026-10-31/` and `v3/lab-mac3153-2026-10-31/`.

4.20r6

The real `Graal.app` 4.20r6 (upload sha256 e1b60bcc…1d9d, executable unmodified) ran on Mac OS X 10.4 Tiger (retail DVD, build 8A428, archive.org `macosx10.4tigerretaildvd`, MD5 28b3352b8b125469f23ed9c3e386f4a3, matches archive.org) in `qemu-system-ppc 8.2.2 -M mac99,via=pmu`, in the cloud container. Server: `seihai-lab/servers/seihai-lab-servers.tar.gz` (GServer-v2 efb5b90)

started with `./run-lab.sh stock`, `G3D27076` added to `allowedversions.txt`, the gserver's `listport` and the lab listserver moved to 14899. `scripts/warpproxy.py` (the 6.x lab proxy, warp host changed to the qemu host's address as seen from the guest) listened on the login server port; the guest's `/etc/hosts` pointed the built-in login server names at the qemu host, as in the 4.04r2 Wine lab. The lab account, throwaway password.

Login line (captured, `run2.pcap`, `proxy.log`)

```
%  <key>  G3D27076  <GSTR account>  <GSTR password>  mac,"",a87b712247c6c9068448ba8c2f68a24b,""\n
```

- id 5, then the key byte, the token `G3D27076`, account and password as GSTR, then the comma list.
- Four comma fields: `mac`, `""`, a 32-hex MD5, `""`. The same line was sent on both connections (Login server and after the 178 warp).
- Field 2 (the binary's hash of the primary Ethernet MAC from IOKit) came out **empty**, as `""`. The guest has a qemu `sungem` NIC with a MAC, so the client's IOKit lookup found nothing it accepts there. Whether a real Mac fills it is not shown by this run.
- Field 3 (hash of `stat("/etc")`) is a plain lowercase 32-hex string.
- Field 4 is `""`, as the static read said.
- Seihai's 4.20r6 row sends `mac,"","",""`. That matches the shape and the empty fourth field. The real client fills field 3.

Client sends seen (C→S ids)

Session 1 (Login server, before the warp) and session 2 (after the 178 warp to `0,Lab Server,HOST,PORT`):

id	what
37	`English`
23	`basepackage.gupd` twice, `bigmap.txt` (session 1 only), then gani/image/font names (`+.gani`, `wbomb1.png`, `wbowi1.png`, `chest.png`, `helveticabd.ttf`, `helvetica.ttf`, `arialbd.ttf`, `state.png`, `sword1.png`, `idle.gani`, `steps.wav`, `sword.gani`)
157	`+`, `idle`, `sword` (gani change)
46	`!`
44	empty (answer to 182)
2	props: first `00 "" / 18=20 / 31=0 / 33=0 / 52 "English" / 75 "mac" / 76=0` (decoded with GServer-v2's prop sizes; raw in `proxy.log`); walking **78/79 (X2/Y2)**, prop 10 gani `walk`/`idle`/`sword`, chat prop 12
34	`&w6\x87$pics1.png`, `&w6\x87#defaulttextedit.png`

id	what
24	) repeated

So the ids seen are 2, 23, 24, 34, 37, 44, 46, 157. Against `4.04-4.20.toml`'s 4.20r6 `pli_sent` (4.20r8's list plus 162), **34 and 46 are missing from that list** but the real client sends both. Seihai's chain already takes 34 and 46 as sent (from 3.153), so Seihai's set covers what was seen.

Not seen: id 18 (flag set). Seihai's 4.20r6 lab run sent `18 gr.x=30` / `gr.y=30.5` from the lab's `gr_movement` weapon (`setstring gr.x,…`); the real client ran the same level and weapon and sent no 18 in either session.

Window title stayed empty; the level, HUD (A/S/D, AP) and chat text (white with a blue shadow) drew. The player sprite did not draw: this server pack has no body/head art (the 4.04r2 lab copied 2.22 `graal_reborn` art for that).

Server ids received (session 2)

4, 6, 7, 9, 10, 14, 25, 30, 33, 34, 39, 41, 42, 43, 47, 82, 100, 101, 168, 182, 190, 194. The client stayed connected through all of them, including 41 (no case in the static read).

Notes and caveats

- **qemu has no 3D**: the client asks AGL for `RGBA, DOUBLEBUFFER, ACCELERATED (0x49), NO_RECOVERY (0x48), DEPTH 32/16` (static: the engine's pixel-format setup in the Mac binary) and stops with "Could not find valid pixel format" (`console-unshimmed-run.log`). `aglshim.c`, built in the guest with the DVD's gcc 4.0, is injected with `DYLD_INSERT_LIBRARIES` and drops only those two attributes, so Apple's software renderer is used. The executable and its signature files are untouched.
- First try was on the beta4 pack: GServer-v2 beta4 **segfaulted** (strlen on NULL in libc) on this login (`proxy-run1-beta4-crash.log`, `run1-beta4-crash.pcap`). The efb5b90 stock pack accepted it.
- One stray chat line (`curl …`, typed into the game by mistake) is in `proxy.log` at the end; ignore it.
- `scripts/` has the qemu driving scripts (monitor screendump, closed-loop mouse for Tiger's relative USB mouse, typing).

3.153

The real Mac `Graal.app` 3.153 (from the `GraalSetup.dmg` upload, executable sha256 6eb0a1f9…83fc, unmodified) ran on the same Tiger 10.4 qemu guest as the 4.20r6 run (the 4.20r6 section above), with the same `aglshim.dylib` (drops only AGL_ACCELERATED and AGL_NO_RECOVERY so the software renderer is used). Servers: `seihai-lab-servers.tar.gz` (GServer-v2 efb5b90) with `./run-lab.sh stock`, `G3D28015` added to `allowedversions.txt`. The lab listserver (`listserver-lab-qemu.py`) is the pack's, with two lab edits: its gserver port moved, and the server entry's address is given as `HOST:PORT` of the qemu host and the logging proxy in place of the loopback address. The guest's `/etc/hosts` pointed the built-in listserver names at the qemu host. `warpproxy.py` ran with `NOWARP=1` in front of the gserver (no 178 injection). The lab account, throwaway password.

Listserver

`listserver.log`: one connection to 14922, id 7 in gen 5 with token `G3D30123` and list `newmain`, then id 1 with the account. The client showed the server list (tab "Hosted") and connected from it.

Game login (captured)

First frame (plain zlib), four messages in one frame:

```
5   %<key>G3D28015<GSTR account><GSTR password>
37  English
34  #\x9dyO$bigmap.txt     34  #\x9dyO$minimap.txt     (then both again)
```

No platform/machine comma list on the login line (unlike 4.x). Same layout as the 3.04 Linux lab capture, with the token `G3D28015`.

First props (id 2): `0 "unknown"` (the start window's nickname field), `18=20`, `31=0`, `33=0`, `52 "English"`, **`75 "mac"`**, **`76 = 1250`** (GINT bytes `20 29 82`). Seihai's 3.153 row took OSTYPE mac from the binary and the code page 1250 from 3.04; both are now seen on the wire.

Walking: X2/Y2 confirmed

Walking sends props **78 (X2) and 79 (Y2)**, never 15/16 (old X/Y):

```
2  *$walk1#n'p          (gani walk, sprite/dir, X2)
2  n'\x80  n'\x90 ...    (X2 every ~150 ms)
2  *$walk1"o)0  o)@ ...  (Y2)
2  *$idle
```

So Seihai PR #294 (3.153 profile to X2/Y2) is right.

Non-`client.` flags

The lab level has the `gr_movement` weapon (`//#CLIENTSIDE … setstring gr.x,…; setstring gr.y,…; timeout=0.05`). In this session the client sent **no id 18 at all** (0 in `proxy.log`), while walking for several seconds. The 3.04 Linux lab run did send 18 for this script, and Seihai's 4.20r6 lab run sent `18 gr.x=30`. The real 4.20r6 Mac client also sent none. What this run cannot tell apart: 3.153 not sending non-`client.` flags, or the script not setting them in this build. The real client shows the same nothing either way.

Other sends

C→S ids seen: 2, 23, 24, 34, 46, 157 (plus 5 and 37 in the first frame). - 34 UPDATEFILE for `wbomb1.png`, `wbowi1.png`, `defaultbutton.png`, `playerlist_icons.png`, `walk.gani`, `sword.gani`, `steps.wav`, `steps2.wav`, `sword.wav` (3.x uses 34 where 4.20r6 used 23 for ganis and sounds). - 157 with the gani name (`walk`, `sword`), 46 `!`, 23 `chest.png` and others. - 24 `)` about every 16 to 18 s (614.06, 632.99, 650.49, 666.74, 683.90). - Sword: 34 `sword.gani`, 157 `sword`, prop 10 `%sword`, 34 `sword.wav`, then `idle` 0.75 s later. - Chat: prop 12 (`,/hello from 3153`), drawn white with a blue shadow above the head.

Server ids received: 4, 6, 7, 9, 10, 14, 25, 30, 33, 34, 39, 41, 42, 43, 47, 82, 100, 101, 168, 182, 190, 194. The player sprite drew (3.153 ships its own body/head art).

`version-profiles/lab-v5-2026-10-30.md`

Lab run: real 5.00r7, 5.12 and 5.24r4 clients on GServer-v2

Date: 2026-10-30

Three real Graal 5 Windows clients were run under Wine 9.0 on Xvfb, against the prebuilt GServer-v2 lab server, in a network namespace with loopback only. The clients were the full 5.00r7 install (with the private folders left out), the original MoleBox-packed 5.12 setup executable, and the MoleBox-packed 5.24r4 executable. A logging proxy between client and server decoded gen 5 in both directions. A second version of the proxy could also rewrite or add server messages.

Tags: **[cap]** means seen on the wire or on screen in this run. **[bin]** means read from the client binary. Times come from the proxy log, which has millisecond timestamps. A tick is 50 ms.

Setup, for repeating it

- In the namespace, `/etc/hosts` points `<login-host-4x>*` and `<list-host-3x>*` at the loopback address. The stock `license/license.graal` (host `<login-host-4x>`, port 14900) then reaches the proxy on 14900, which forwards to GServer on 14901.
    - The 5.00r7 folder came with a `license.graal` that had been edited to point at a private LAN host. The stock file from the 5.12 bundle replaced it.
- GServer-v2 needs the tokens in `config/allowedversions.txt`: `G3D22067` (5.00r7), `G3D14097` (5.12), `G3D15010` (5.24r3/r4), `G3D26090` (5.30r3/5.31r4).
- **Stock gs2lib does not know `G3D15010`.** It treats 5.24 as a pre-2.1 client, so file messages come in the old layout. 5.24 then tries to save PNG data as the file name ("Couldn't open …1.png‰PNG… for writing"). In a lab copy of `libgs2lib.so`, the `G3D26090` table entry was renamed to `G3D15010`. With that copy, 5.24r4 downloads files normally. **[cap]** Since 2026-10-31 the proper fix is `tools/lab-servers/gs2lib-CLVER_5_24.patch`, which gives 5.24 its own slot and leaves `G3D26090` in place. The rule (layout chosen by the server's token table) is in `protocol/file-transfer.md`.
- Wine:
    - Under Wine, 5.00r7 and 5.12 draw through Direct3D 7, and 5.24r4 draws through OpenGL. **[cap]**
    - Setting wined3d's renderer to Vulkan gave a black screen. The default GL renderer works. **[cap]**
    - There is no window manager, so `xdotool windowfocus` on the game window is needed before keys reach it. The client also pauses while unfocused. **[cap]**

Login mode: why nothing is drawn at first

- **Fact.** The client treats the server it reaches through `license.graal` as the **Login** server. The console says "Loading tile definitions for server Login", and the window title is " L o g i n ".
- **What it hides.** In Login mode the client draws the level tiles, chests and image NPCs, but no characters (neither its own player nor other players) and no default HUD. On the real service, that server is only the backdrop for the scripted server list. **[cap]**
- **Same check in 4.04.** The v4 lab found the same rule in the 4.04 engine. If the current server name starts with "login" (case-insensitive), the draw-layer mask is 0 instead of the full mask.
- **What did not fix it.** None of these changed anything:
    - adding body, head, sword and shield images and ganis locally;
    - a fresh file-name cache;
    - a gani script reply;

    - a curlevel prop for the player itself;
    - removing the colours prop;
    - a different Wine renderer.
- **The fix is S→C 178 (`PLO_SERVERWARP`).** Send it after the login burst. Here it was sent right after S→C 182, with the text `0,Lab Server,<loopback>,14900` (host field = the loopback address).
    - The client closes the connection and logs in again with a new key, to the host and port in fields 2 and 3.
    - The second connection is named after field 1. The window title becomes " L a b S e r v e r " (5.24r4: "Lab Server").
    - On that connection the client draws:
        - the player, with shadow, head, body, sword and shield;
        - other players, with their nicknames;
        - hearts, the A/S/D item boxes, the counters, the AP bar;
        - the native player list window. **[cap]**
- **GServer-v2 and 178.** GServer-v2 sends 178 only when a listserver answers its server-info request, so the lab proxy injected it. The same warp works for 5.31r4 and 6.03r7 (the "Play v5 and v6 clients" lab).
- **Tile definitions after the warp.** The client asks for `tiledefs<servername>.txt`, with the space escaped as `%032`.
- **Implication for a server emulator.** A v5+ client must end up on a server whose name does not start with "login". Otherwise the player never appears.

Message flow seen in this run

- **Login.** The login frame and the order of the server's login burst are as described in `protocol/v5x-clients.md`. **[cap]**
- **Frame types the client accepts.** The rewriting proxy re-encoded every server frame. With every frame sent as zlib (type 4), the client disconnected after a few frames. It stayed connected once payloads of 55 bytes or less went as type 2 (plain, XOR over up to 12 dwords) and larger ones as zlib. Payloads over 8 KiB went as bz2 (type 6). That 55-byte limit is the one GServer uses. **[cap]**
    - Only the combination was tested. The exact rule the client applies is not known.
- **Movement.**
    - While walking, the client sends a props message (C→S 2) **every tick**. It carries only the coordinate that changed: X2 (prop 78) or Y2 (prop 79). The value is pixels × 2, with the low bit as the sign, so positions go out in whole pixels.
    - The gani change goes in the same message as the first step (`*` walk, then `n/o`).
    - 5.24r4 also adds prop 17 (sprite/direction) to gani changes. 5.00r7 and 5.12 do not. **[cap]**
- **Files.**
    - A gani is checked with C→S 34 (UPDATEFILE), or with C→S 23 (WANTFILE) when it is not cached.
    - Then comes C→S 157 (the gani name and a 5-byte checksum). The server answers with S→C 195 (gani name and attribute text, for example `SETBACKTO idle` for `sword`). **[cap]**
    - 5.12 and 5.24r4 fetch `sword.wav` and `lift.wav` from the server with WANTFILE the first time they are needed. 5.00r7 has them locally. **[cap]**
- **Pushing into an object.** While the push pose lasts, the client sends C→S 38 with the text `pushed,0` and the object's position. The first goes about 550 ms after `push` starts, then one every second. **[cap]**
- **Cutting and lifting a bush.**
    - Both send C→S 1 (BOARDMODIFY) with the 2×2 stump tiles. GServer echoes it to everyone as S→C 7. **[cap]**
    - Lifting sets prop 19 (carry sprite) to 1.

    - Throwing sends C→S 11 (no payload) and sets the carry sprite to −1. **[cap]**
- **Respawn is the server's job.** GServer-v2 restored the bush with S→C 7 exactly 15.0 s after the cut, which is its `respawntime = 15`. The client does not put tiles back by itself online. The respawn time a real 5.x server used was not observed. **[cap]**
- **Periodic messages.** C→S 24 about every 5 s, and S→C 42 (world time) every 5 s. **[cap]**
- **Clientside scripts.** NPC scripts arrive as source text in S→C 3 (with `§` for line breaks), not as bytecode. Clientside NPC scripts were only tried while the clients were still in Login mode, so whether they run was not settled. **[cap]**

Timings

All three builds gave the same numbers.

action	5.00r7	5.12	5.24r4
walk	0.5 tile (8 px) per tick, 41 updates in 2.0 s	same	same
first step after key down	about 1 tick (16–54 ms)	same	same
key released → `idle`	next tick (38 ms)	same	same
last refused step → `push`	908 ms (18 ticks)	902 ms	902 ms
`sword` → `idle`	248 ms (5 ticks)	243 ms	246 ms
`lift` → `carrystill`	304 ms (6 ticks)	not run (bush already cut)	299 ms
carry walk	0.5 tile per tick, `carry`	—	same
hidden walk (D while carrying)	—	—	35 px in 22 ticks = 0.1 tile per tick, `hidden` / `hiddenstill`
blocked against a corner	small x nudges of 1–2 px per tick toward the gap	—	—

These match the 2.05 timings in `client-internals/player-mannerisms-live.md`: push on the 18th blocked tick, sword 5 ticks, lift 6 ticks, hidden 0.1 tile per tick. **[cap]**

How it looks

- **Chat text** (another player's chat, seen from 5.12):
    - It is drawn above the head, centred, as white bold sans-serif text with a blue drop shadow down and to the right.
    - The shadow colour is mostly RGB (88, 88, 255), with some pure (0, 0, 255).
    - There is **no bubble box**. **[cap]**
    - The nickname is drawn the same way under the player.
    - A player whose window is inactive shows as `name (paused)`.
    - The client's own nickname is not drawn under its own player.
    - Shot: `client-internals/mannerisms-shots/v5-512-chat-text-zoom.png`.
- **Chat input needs the server.** When it decides whether keyboard input belongs to a text field, the engine looks for a GUI control named `ChatBar`. **[bin]**
    - On a bare GServer-v2 there is no such control. Typing, Tab and Enter do nothing, and no chat message is sent. **[cap]**
    - The chat bar is therefore part of the server's scripted GUI, not of the engine. The chat text above was injected as prop 12 through the proxy.
- **Carrying and hiding:**
    - A carried bush is drawn over the top of the head. Facing down, the face stays visible below it (`v5-500r7-carry-down.png`).

    - In `hiddenstill`/`hidden` only the bush is drawn, at the player's feet, and the player is not drawn at all (`v5-524-hold-hide-hidden.png`). **[cap]**
- **HUD:**
    - 5.00r7 and 5.12 label the item boxes A, S and D and print the sword level next to the sword.
    - 5.24r4 draws the same boxes without the letters and without the level number (`v5-hud-512-vs-524.png`).
    - **Possibly Wine-specific.** 5.24r4 is the one build that draws through OpenGL here. It may simply not be drawing that text under Wine, so confirm this on real Windows before using it in a profile. **[cap]**
- **Player list.** The player list is a native Windows window, titled "Players: N online", with tabs and the groups Staff, Buddies, Guild, Normal, Ignore and Offline. 5.00r7 and 5.12 opened it at login. 5.24r4 did not. **[cap]**
- **Tile graphics.**
    - In Login mode the level used one tile set.
    - After the warp, 5.00r7 used its newer default tile graphics. Bushes became the red-and-white flower bushes, and stones became pale green blocks (`v5-500r7-after-warp.png`).
    - The client asks for `tiledefs<servername>.txt` after the warp, so the change of art probably comes from tile definitions chosen per server name. **[cap]** for the request, the cause is an inference.

Not covered here

- Pots and stones: the test level has none. The respawn rule above (the server restores the tiles after its `respawntime`) applies to them as well; their per-object timing was not measured.
- 5.30r3 and 5.31r4 were run by the "Play v5 and v6 clients" lab, not here.

Lab files

The project files hold the lab tools and the raw logs, under `graal-decompile/v5x/lab/`:

- the proxy and the rewriting proxy (`g5mitm.py` with `rules.py`);
- the input driver and the measurement suite (`drive.py`, `suite.sh`, `analyse.py`);
- the raw proxy logs and the full-size screenshots.

`version-profiles/matrix/builds-deltas.md`

Per-build deltas from the static survey, 1998 to 2.31

Date: 2026-10-30. Generated from the survey in `README.md` (method there). Each line lists what a build adds (+) or drops (−) against the build listed before it, in link order. Graal3D is a separate program and is listed on its own. **[bin]** strings and form resources; **[file]** shipped files.

`graalconfig.txt` keys

Plain `key=` strings. From 1.38 some keys are built one character at a time, so a key can be missing from this list while the build still reads it; garbled three-letter keys (`ekv`, `u5j`, `d4n`, `kxs`, `azg`) are such fragments.

- **6th Graal (beta5)**: + `joystickon=`, `musicon=`, `reversestereo=`, `soundfxon=`, `waitvertical=`
- **8th Graal (beta8)**: + `saveimages=`
- **1.1 (15th)**: + `editnickname=`, `gamemode=`
- **1.21**: + `accountname=`, `loadcolorsfromserver=`, `password=`, `serveraddr=`, `serverport=`

- **1.28 beta2**: + `compabilitydraw=`, `externalchat=`, `language=`, `nodecompressionerror=`
- **1.3 test**: + `font=`, `fontsize=`; − `accountname=`, `password=`
- **1.31**: + `fixedport=`, `fixedudpport=`, `noudp=`; − `editnickname=`
- **1.32**: + `nomassmessages=`
- **Graal3D** (separate program): `bodytex=`, `bodyx=`, `fixedport=`, `fixedudpport=`, `headx=`, `noudp=`, `serveraddr=`, `serverport=`
- **1.33**: + `nodirectinput=`
- **1.35**: + `nohtmlinpms=`, `u5j=`
- **1.36**: + `limitnicknames=`, `nicknamelimit=`; − `serverport=`
- **1.38**: − `externalchat=`
- **1.39**: + `dontconnectlevels=`, `dontloadlistheads=`, `nohtmlimages=`, `notoalls=`, `sortplayersbyname=`
- **1.41 rev1**: + `dontsavepasswords=`, `fullscreenmode=`, `usewinamp=`; − `serveraddr=`
- **NewMain 1.42 (24)**: + `dontsavepms=`, `dontusedirectx7=`, `ekv=`, `head=`, `joyselected=`; − `gamemode=`
- **2.00**: − `loadcolorsfromserver=`
- **NewWorld beta 1**: − `ekv=`
- **2.05 rev2**: + `ekv=`
- **2.1 test 2**: + `startinfullscreen=`; − `ekv=`
- **2.17.1**: + `automapping=`, `d4n=`, `kxs=`, `microactivatebyvolume=`, `microactivatekey=`, `microactivationlevel=`, `microinputdevice=`, `microon=`, `midivolume=`, `mp3volume=`, `quickplayerworlds=`, `radiovolume=`, `sfxvolume=`, `voicecodec=`, `voicevolume=`; − `musicon=`, `soundfxon=`, `usewinamp=`
- **2.22**: + `azg=`; − `d4n=`, `kxs=`
- **2.31**: + `cachefilenames=`

Sound effect names (`.wav`)

Names the executable refers to. Sounds named only by scripts or ganis are not here.

- **4th Graal alpha**: + `arrow.wav`, `axe.wav`, `bomb.wav`, `chest.wav`, `crush.wav`, `dead.wav`, `extra.wav`, `frog.wav`, `lift.wav`, `put.wav`, `sword.wav`, `swordon.wav`
- **7th Graal (beta7)**: + `water.wav`
- **8th Graal (beta8)**: + `arrowon.wav`, `compudead.wav`, `fire.wav`, `goal.wav`, `horse.wav`, `horse2.wav`, `item.wav`, `item2.wav`, `jump.wav`, `lift2.wav`, `nextpage.wav`, `sign.wav`, `steps.wav`, `steps2.wav`, `stonemove.wav`, `thunder.wav`
- **1.1.9 Gold**: − `horse.wav`, `horse2.wav`, `steps2.wav`
- **Graal3D** (separate program): none
- **1.34**: + `horse2.wav`
- **NewMain 1.42 (24)**: − `dead.wav`, `goal.wav`, `sign.wav`
- **NewWorld beta 1**: − `chest.wav`, `compudead.wav`, `frog.wav`, `item2.wav`, `jump.wav`, `stonemove.wav`, `swordon.wav`
- **2.05 rev2**: + `chest.wav`, `compudead.wav`, `frog.wav`, `item2.wav`, `jump.wav`, `stonemove.wav`, `swordon.wav`
- **NewWorld beta 2**: − `chest.wav`, `compudead.wav`, `frog.wav`, `item2.wav`, `jump.wav`, `stonemove.wav`, `swordon.wav`
- **2.17.1**: + `chest.wav`, `compudead.wav`, `frog.wav`, `item2.wav`, `jump.wav`, `stonemove.wav`, `swordon.wav`

Image names the executable refers to

- **1st Graal**: + `letters.gif`, `link.gif`, `map.gif`, `opps.gif`, `pics1.gif`, `state.gif`
- **3rd Graal**: + `heads.gif`
- **5th Graal**: + `sprites.gif`; − `link.gif`
- **5th Graal alpha**: + `map2.gif`
- **9th Graal alpha**: + `graal.bmp`, `ride.gif`, `ride2.gif`, `servant1.gif`
- **9th Graal (beta9)**: + `w-warpdoor.gif`
- **1.0 (11th)**: + `warpdoor.gif`; − `w-warpdoor.gif`
- **1.1 (15th)**: + `map3.gif`
- **1.21**: + `baddyblue.gif`, `baddydragon.gif`, `baddygold.gif`, `baddygray.gif`, `baddyhare.gif`, `baddylizardon.gif`, `baddyoctopus.gif`, `baddyred.gif`, `head0.gif`, `head1.gif`, `head2.gif`, `head3.gif`, `shield1.gif`, `shield2.gif`, `shield3.gif`, `sword1.gif`, `sword2.gif`, `sword3.gif`, `sword4.gif`, `wbomb1.gif`, `wbomb2.gif`, `wbomb3.gif`, `wbow1.gif`, `wbow2.gif`, `wbow4.gif`, `wbowi1.gif`, `wbowi2.gif`, `wbowi3.gif`, `wbowi4.gif`; − `heads.gif`, `ride.gif`, `ride2.gif`, `servant1.gif`, `warpdoor.gif`
- **1.28 beta2**: − `map3.gif`
- **1.3 test**: + `ss_arrow.gif`, `ss_bomb.gif`
- **Graal3D** (separate program): `body.bmp`, `logo.bmp`, `man.bmp`
- **1.33**: + `body.png`, `bomb1.png`, `bomb2.png`, `letters.png`, `map.png`, `map2.png`, `opps.png`, `pics1.png`, `sprites.png`, `state.png`; − `letters.gif`, `map.gif`, `map2.gif`, `opps.gif`, `pics1.gif`, `sprites.gif`, `state.gif`
- **1.34**: + `graal.png`
- **1.35**: + `graalmap.bmp`; − `head0.gif`, `head1.gif`, `head2.gif`, `head3.gif`
- **1.41 rev1**: + `graalmap.png`; − `graal.bmp`, `graalmap.bmp`
- **NewMain 1.42 (24)**: + `head0.gif`
- **2.00**: + `1hlight2.png`, `ahlight2.png`, `arrowsbox.gif`, `bluelampani.gif`, `door.gif`, `gate1.gif`, `gate2.gif`, `head19.gif`, `head24.gif`, `head26.gif`, `head27.gif`, `head495.gif`, `light2.png`, `sleepkoji.gif`, `tutorial_housegate.png`, `tutorialmap.png`, `tutorialmap2.png`
- **NewWorld beta 1**: + `bodynw.png`, `headalgo000.gif`, `newsword.png`, `pics1nw.png`, `spritesnw.png`; − `1hlight2.png`, `ahlight2.png`, `arrowsbox.gif`, `bluelampani.gif`, `body.png`, `door.gif`, `gate1.gif`, `gate2.gif`, `head19.gif`, `head24.gif`, `head26.gif`, `head27.gif`, `head495.gif`, `light2.png`, `sleepkoji.gif`, `sprites.png`, `tutorial_housegate.png`, `tutorialmap.png`, `tutorialmap2.png`, `wbow1.gif`, `wbow2.gif`, `wbow4.gif`
- **2.05 rev2**: + `1hlight2.png`, `ahlight2.png`, `arrowsbox.gif`, `bluelampani.gif`, `body.png`, `door.gif`, `gate1.gif`, `gate2.gif`, `head19.gif`, `head24.gif`, `head26.gif`, `head27.gif`, `head495.gif`, `light2.png`, `sleepkoji.gif`, `sprites.png`, `tutorial_housegate.png`, `tutorialmap.png`, `tutorialmap2.png`, `wbow1.gif`, `wbow2.gif`, `wbow4.gif`; − `bodynw.png`, `headalgo000.gif`, `newsword.png`, `pics1nw.png`, `spritesnw.png`
- **NewWorld beta 2**: + `bodynw.png`, `headalgo000.gif`, `newsword.png`, `pics1nw.png`, `spritesnw.png`; − `1hlight2.png`, `ahlight2.png`, `arrowsbox.gif`, `bluelampani.gif`, `body.png`, `door.gif`, `gate1.gif`, `gate2.gif`, `head19.gif`, `head24.gif`, `head26.gif`, `head27.gif`, `head495.gif`, `light2.png`, `sleepkoji.gif`, `sprites.png`, `tutorial_housegate.png`, `tutorialmap.png`, `tutorialmap2.png`, `wbow1.gif`, `wbow2.gif`, `wbow4.gif`
- **2.17.1**: + `baddyblue.png`, `baddydragon.png`, `baddygold.png`, `baddygray.png`, `baddyhare.png`, `baddylizardon.png`, `baddyoctopus.png`, `baddyred.png`, `body.png`, `bshield0.png`, `emoticonmicro.png`, `g2k2pet_squirrelm0.png`, `graal2002letters.png`, `head0.png`, `head26.png`, `khairs0.png`, `khead0.png`, `klegs0.png`, `kmarms100.png`, `kmbody100.png`, `picso.png`, `shield1.png`, `shield2.png`, `shield3.png`, `sprites.png`, `ss_arrow.png`, `ss_bomb.png`, `sword1.png`, `sword2.png`,

`sword3.png`, `sword4.png`, `wbomb1.png`, `wbomb2.png`, `wbomb3.png`, `wbow1.png`, `wbow2.png`, `wbow4.png`, `wbowi1.png`, `wbowi2.png`, `wbowi3.png`, `wbowi4.png`; − `baddyblue.gif`, `baddydragon.gif`, `baddygold.gif`, `baddygray.gif`, `baddyhare.gif`, `baddylizardon.gif`, `baddyoctopus.gif`, `baddyred.gif`, `bodynw.png`, `head0.gif`, `headalgo000.gif`, `newsword.png`, `pics1nw.png`, `shield2.gif`, `shield3.gif`, `spritesnw.png`, `ss_arrow.gif`, `ss_bomb.gif`, `sword1.gif`, `sword2.gif`, `sword3.gif`, `sword4.gif`, `wbomb1.gif`, `wbomb2.gif`, `wbomb3.gif`, `wbowi1.gif`, `wbowi2.gif`, `wbowi3.gif`, `wbowi4.gif`

- **2.31**: − `graalmap.png`, `shield1.gif`, `ss_arrow.png`, `ss_bomb.png`

Delphi forms (windows)

Form class names from the RCDATA resources.

- **1st Graal**: + `TADVENTURE`, `TCONNECTFRAME`, `TEDITBADDY`, `TEDITSIGNS`, `TLINKLIST`, `TLINKOPTIONS`, `TLOADLEVEL`, `TOBJECTNAME`, `TOPTIONS`, `TSHOWCODES`
- **2nd Graal**: + `TCHESTEDIT`, `TEDITPERSON`, `TPERSONACTION`
- **3rd Graal**: + `TEDITFLAGS`
- **5th Graal**: + `TMOREOPTIONS`
- **6th Graal (beta5)**: + `TOPENURL`
- **9th Graal alpha**: + `TRECONNECTFRAME`
- **1.1 (15th)**: + `TCONTROLSFRAME`
- **1.21**: + `TDEBUGGERFRAME`
- **1.28 beta2**: + `TEDITCHARACTER`, `TMESSAGEEDITOR`, `TPLAYERLIST`
- **1.32**: + `TRTFMESSAGE`
- **Graal3D** (separate program): `TADVENTURE`, `TCONNECTFRAME`, `TEDITLOOK`, `TMOREOPTIONS`
- **1.39**: + `TPLAYERLISTOPTIONS`, `TPROFILEEDIT`
- **1.41 rev1**: + `TSERVERLISTFRAME`, `TSERVERMESSAGE`, `TSHORTMESSAGE`, `TWAABOUT`; − `TRTFMESSAGE`
- **2.04**: + `TOBJECTSET`
- **NewWorld beta 1**: − `TCHESTEDIT`, `TEDITBADDY`
- **2.05 rev2**: + `TCHESTEDIT`, `TEDITBADDY`
- **2.1 test 2**: + `TEDITNPC`; − `TEDITPERSON`, `TOBJECTSET`, `TPERSONACTION`, `TSHOWCODES`
- **NewWorld beta 2**: − `TCHESTEDIT`, `TEDITBADDY`
- **2.17.1**: + `TABOUTDLG`, `TCHESTEDIT`, `TEDITBADDY`, `TLOGFRAME`, `TRPGFRAME`, `TTILESARRAYFRAME`; − `TWAABOUT`
- **2.31**: − `TCHESTEDIT`, `TDEBUGGERFRAME`, `TEDITBADDY`, `TEDITCHARACTER`, `TEDITFLAGS`, `TEDITNPC`, `TEDITSIGNS`, `TLINKLIST`, `TLINKOPTIONS`, `TOBJECTNAME`, `TTILESARRAYFRAME`

Font names in the executable

- **1st Graal**: + `Courier`, `MS Sans Serif`, `System`
- **6th Graal (beta5)**: − `Courier`
- **Graal3D** (separate program): `MS Sans Serif`, `System`
- **1.34**: + `Courier`, `Courier New`
- **2.17.1**: + `Arial`

`version-profiles/matrix/live-runs-2026-10-30.md`

Live runs of six eras under Wine, 2026-10-30

Six clients were run offline in a cloud container: 1st Graal (Sep 1998), 1.0 (10th Graal, Dec 1998), 1.1 (15th Graal, Jan 1999), 1.32 (Jan 2000), 1.41 rev1 (Nov 2000) and 2.00 (Feb 2001). Each was started in its own single-player or training mode, screenshotted, and timed while walking. No network: every client ran inside `unshare --net`. **[live]**

The screenshots, the helper scripts and the survey data are in the project folder under `graal-decompile/version-matrix/` (`shots/`, `tools/`, `survey.jsonl`). They are not in this repo.

Setup notes for the old builds

These matter for anyone running a pre-2001 client in the lab (`../../tools/lab-setup.md`, and the live-trace setup used for 2.05):

- **Keyboard.** The 1998–1.41 clients read the keyboard through DelphiX `TDXInput`, which loads DirectInput. Under Wine 9.0 no key reaches them that way. With `WINEDLLOVERRIDES="dinput=d"` the DirectInput load fails, DelphiX falls back to the Windows keyboard state, and `xdotool keydown` works.
- **A window manager.** With bare Xvfb the game window never becomes active. With `openbox` running and the Wine virtual desktop on (`HKCU\Software\Wine\Explorer Desktop=Default`, `Desktops\Default=1024x768`), the window takes focus normally.
- **Idle-driven clients.** From NewMain 1.42 the frame runs from the idle handler. Under Wine it only advances while window messages arrive, so a screen can look frozen until the mouse moves. Keep posting small mouse moves during a measurement.
- **Missing files.** 2.00's archive holds only the exe. It runs from the 2.22 install folder once `objects.dat` (from any 1.x install) is copied in. The 1.21 InstallShield groups need their heads, swords and shields placed under `levels/` before the start window will open.
- **Reading positions.** The player position is a pair of doubles in tiles in every build tried. They were found by scanning for values that rise while Right (or Down) is held and fall while Left (or Up) is held, then polled every 2 ms with unbuffered reads of `/proc/<pid>/mem`. (A buffered reader returns stale bytes and makes the game look frozen.)

Walking speed

Build	Step	Interval	Speed	Notes
1st Graal	0.5 tile	50 ms	10.0 tiles/s	x from 12.5 to 14.0 in three steps
1.0 (10th)	0.5	50	10.0	
1.1 (15th)	0.5	50	10.0	After getting out of bed and bumping the furniture, x carried a 0.025 offset (26.925) and kept it while walking. 2.05's wall rule produces such offsets (`x + 0.525`-style probe points and 0.1 nudges), so a similar rule is likely; not read
1.32	0.5	50	10.0	Measured on y

Build	Step	Interval	Speed	Notes
1.41 rev1	0.5	50	10.0	Measured on y. A second copy of y moves in 0.25 steps (the view or the drawn position)
2.05 rev2	0.5	50	10.0	From the 2026-10-28 live trace (`walkBC`)

So walking speed does not change from the first beta to 2.05, and the 2.17, 2.22 and 2.31 walking routines use the same half-tile step (`../../client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md`).

2.00 was not timed: the training level opens with a scripted scroll that holds the player, and it did not finish within the run.

What the screenshots show

Build	Start window	In game
1st Graal	"Graal - The Adventure"; "Select Game Mode" with One Player, Two Player, One Player Online; server the loopback address, port `14801`; "Graal by Stefan Knorr, Presented by <cyberjoueur-host>"	Bedroom start. Bomb and bow icons, rupees as three digits `000`, bombs `10`, arrows `05`, three spiral hearts
1.0	"Select the game mode", grail picture, first and second player characters (Nick, Mandi), "Online At Cyberjoueur" with port 14802	Same bedroom with the blue frame. Rupees without leading zeros (`0`). The view scrolls with the player
1.1	Rider-on-horse picture, game-mode and character drop-downs (Fung Si Yan)	The player starts asleep in the left bed
1.32	As 1.1 plus Language and Nick Name fields and "Managed by: Stéphane Portha"	A/S/D buttons, the offline start level with NPCs "Stefan (Coder)" and "Kain (GP)", and a scrolling hint line ("If you have problems, then just press F1 and click on 'More options'!"). A chat line sits under the view
1.41	Language, Game mode (One Player, One Player Online, Two/Three/Four Players), Character, Nick Name; "Lead programmer: Stefan Knorr"	As 1.32 plus the AP bar under the A/S/D buttons and a small number beside the sword
2.00	Dark green "Graal 2001" window, GraalOnline logo, Language and Nickname, Training / Options / Start, "Copyright © 1998–2000 Cyberjoueur"	Training level with a parchment scroll ("Welcome to Graal!"), light effects, a minimap at the bottom left, name "Stefan" in a smaller rounded face

Under Wine the player body is sometimes not drawn (2.00) and stale strips stay after a scroll, as noted for 2.05 in `../../client-internals/live-trace-205.md`. Trust the memory reads over the pictures for anything that moves.

`version-profiles/matrix/README.md`

Per-version profile matrix, 1998 to 2014

Date: 2026-10-30

This is the one-page view of how each Graal client **thinks** (protocol, scripting), **acts** (timing, movement, rules, sound) and **looks** (window, start screen, HUD, fonts, menus), from the Java Zelda Online of 1998 to the 6.11 Mac client of 2014, plus the Graal3D side project. Each cell is short and points at the note that has the detail. It is meant as the checklist for a client that wants to *be* a chosen version, such as Seihai's version selector.

New in this pass (2026-10-30):

- A static survey of all 57 Delphi client executables (1st Graal to 2.31): forms, main window, frame timer, fonts, sound names, config keys and shipped sheets. Per-build values are in `builds-static-survey.tsv`, and what each build adds or drops is in `builds-deltas.md`.
- Live runs under Wine of 1st Graal, 1.0, 1.1, 1.32, 1.41 and 2.00, with screenshots and a walking-speed measurement (`live-runs-2026-10-30.md`).
- A comparison with Seihai's per-version profiles (`seihai-gaps.md`).

Tags, as elsewhere in this repo: **[bin]** read from the client binary · **[file]** shipped files · **[live]** seen in a running client · **[cap]** packet capture · **[srv]** server source · **[doc]** release notes · **[inf]** inferred.

Eras

The builds fall into eleven eras. Inside an era the clients share their engine and wire format, so a profile can be one era row plus the per-build deltas listed in the linked notes.

Era	Builds	Dates	Engine	Detail
Z	Zelda Online 1 and 2 (Java applets), GServ Standalone	May–Jul 1998	Java	`../../client-internals/zelda-online-1998.md`, `../../client-internals/precursors-and-fan-builds.md`
A	1st Graal to 1.1.9 Gold (20 builds)	Sep 1998 – Feb 1999	Delphi 2-level compiler (2nd Graal: Delphi 4), DelphiX	`../../client-internals/build-catalogue.md`, `../../protocol/relay-1998.md`
B	1.21, 1.28 beta 2, 1.3 test, 1.31, 1.32	May 1999 – Jan 2000	Delphi 4 from 1.28b2, DelphiX	`../../protocol/gserv-1999.md`
3D0	Graal3D	Feb 2000	Delphi, DelphiX Direct3D retained mode	`../../graal3d/README.md`
C	1.33 to 1.39	Feb – Aug 2000	Delphi 4, DelphiX	`../../client-internals/versions-133-139.md`
D	1.41 rev1	Nov 2000	as C	`../../client-internals/version-profiles/README.md`
E	NewMain 1.42, 2.00 to 2.05 rev2, 2.1 tests, NewWorld betas	Jan – Oct 2001	as C, plus DirectDraw 7	`../../client-internals/cross-version-timeline.md`, `../../client-internals/newworld-betas.md`
F	2.1 to 2.17 (only 2.17.1 survives as a binary)	Oct 2001 – 2002	as E, BASS audio from 2.17 at the latest	`../../client-internals/version-profiles/141-217.toml`
G	2.22, 2.3 (no binary), 2.31	2004 – 2005	as F	`../2.22-2.31.md`
H	3.04 Linux, 3.153 Mac	2004 – 2005	C++ on the Torque Game Engine	`../../protocol/v3x-clients.md`
I	4.04r2, 4.20r6/r8	2006 – 2007	Delphi launcher + C++ `GraalEngine.dll` (+ `Graal3DEngine.dll`)	`../../protocol/v4x-clients.md`

Era	Builds	Dates	Engine	Detail
J	5.00r7, 5.12; then 5.24 to 5.31	2007 – 2011	5.00/5.12 as I; from 5.24 one MSVC exe (`quattro_lib`)	`../../protocol/v5x-clients.md`, `../../protocol/v5v6-windows-unpacked.md`
K	6.00 betas to 6.03r7, 6.11 Mac	2010 – 2014	one native C++ program per platform, no Torque layer	`../../protocol/v6x-clients.md`

Thinks: wire and scripting

Era	Login token	Cipher gen	Server discovery	Ids handled / sent	Props parsed	Scripting
Z	none	text lines	`host:14801`	text lobby lines	–	none
A	none	1 (no zlib)	connect dialog, the loopback address on 14801, Cyberjoueur relay 14802	relay lobby `N/P/C/D` lines, `D` payload types per build	relay `D` types	GS1-like, `this.` vars from 1.1 beta1 **[bin][doc]**
B	`GSERV007–018` (a protocol revision, not a date)	2 (zlib, no cipher)	connect dialog, `<list-host-classic>:14804`, `:14900` from 1.31	GSERV id ranges (`gserv-1999.md`)	–	GS1, NPC weapons from 1.21
3D0	`GR3DV002`	own (2D codec, no key byte)	the loopback address on 14901, UDP 14898 peer to peer	own renumbered ids	3D floats, 24-bit colours	none
C	`GNW` + date (`GNW20020` … `GNW07080`)	2	connect dialog `<list-host-classic>:14900`, guest/guest	1.33: S→C 0–41, 100; C→S 0–31; grows to 45 / 35 ids by 1.39	own 0–33/34, others 21–23 ids	GS1, 88 → 111 commands
D	`GNW13110`	**3** (key byte + LCG insertion)	first server-list window	+1.41 deltas	props to 36	GS1, 126 commands
E	`GNW08011` … `GNW03081`; NW betas `G2N15041`; 2.1 tests keep `GNW03081`	3	listserver, host DES-hidden from 2.00, `:14921` (2.1 tests `:14922`)	2.05: 59 S→C ids	to 42 (2.05), 50, 51	GS1 134 → 161 commands, ganis from 1.42
F	`GNW31101` (2.1) … `GNW22122` (2.17)	3	2.17: signed `arrays2.dat`, id 0 then id 1	2.17: 79 S→C, 61 C→S; TAMPERCHECK (39) from 2.17	to 74 (gani attrs `#P10`–`#P30` from 2.16)	GS1 + gani scripts (2.1), `this.` client strings (2.19)
G	`GNW03014`, `GNW14015`, `GNW28015`; list `GNW30123`	**5** (compression byte, partial XOR, bz2)	listserver id 7 `GNW30123newmain`	2.22: 80, 2.31: 81	to 74 (2.22), 80 with X2/Y2/Z2 (2.31)	GS1; classic client-side NPC mode removed in 2.3
H	`G3D03014`, `G3D28015`; list `G3D30123`	5	RSA-signed `license.graal` → `<list-host-3x>:14922`	79 (3.04), 88 (3.153)	as G	GS1 native + TorqueScript `.dso` for menus; NPC bytecode ids (3.153)
I	`G3D17026`, `G3D27076`, `G3D20126`	5	`license.graal` → `<login-host-4x>:14900`; server named `Login`, no separate list protocol	98–101	–	GS2 bytecode, server-pushed ganis (195), update packages `GRPKG001`

Era	Login token	Cipher gen	Server discovery	Ids handled / sent	Props parsed	Scripting
J	`G3D22067`, `G3D14097`, `G3D15010`, `G3D26090`	5	as I; from 5.24 the hosts are compiled in, no license file	95–100	–	GS2, `GRESCR01` script cache; anti-cheat process scan (182/183)
K	`G3D0108A`/`G3D0511C` (Linux), `G3D0311C` (Win 6.03); 6.11 set by script	5; 6.11 can switch to a **script-defined** header with RC4/AES/TLS	compiled-in `loginserver*:14900`; 6.11 HTTPS connector to `con.quattroplay.com`	91–97	–	GS2; in 6.11 even the packet id tables come from the server's script

Login-time extras that a faithful client must send: `PLI_LANGUAGE` (37) from 2.02 rev2 **[bin]**; `PLI_TAMPERCHECK` (39) once (2.17), on gmap crossings and level links (2.22), or every 3.2 s (2.31) **[cap]**; machine-id MD5s after the password from 4.x on **[bin]**.

Acts: timing, movement and rules

What	Value by era	Source
Game tick	**50 ms** in every Delphi build. A through D drive it from a DelphiX `TDXTimer` (interval 50 in the main form); from NewMain 1.42 the form has no DX timer and the frame runs from the idle handler behind a 50 ms gate. Graal3D: 50 ms `TDXTimer`.	**[bin]** survey; `../../client-internals/rendering-observations.md`
Walking	**0.5 tile per tick, 10 tiles per second**, from the first beta on. Measured live in 1st Graal, 1.0, 1.1, 1.32, 1.41; the 2.05 trace and the 2.05/2.17/2.22/2.31 walking routine agree.	**[live]** `live-runs-2026-10-30.md`; `../../client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md`
Wall test	**1st Graal:** six probes, fail closed, push counter wraps above 37 back to 18, hop at 18 (`rules-1st-graal.md`). **1.39 and 2.05–2.31:** six probe points per facing step. 2.05 also has corner nudge 0.1 and flush snap. In 1.1 a position with a 0.025 offset (x = 26.925) showed up after bumping furniture, which fits a similar rule.	**[bin]** 1st Graal, 1.39 (`rules-139.md`), 2.x; **[live]** 1.1
Death and respawn	**1st Graal:** hearts to 0 set the dead life-state; a key once the shared counter is above 27 restarts `house1.graal` at (24.5, 30) with 3 hearts. The up-count was not found. **1.39:** 30-tick timer, input and STATUS dead bit after counter > 27, online default (30.5, 30), hearts restored to max (no AP bands). **2.05:** same timer/bit; hearts by AP; 2.22 lab-confirmed.	`rules-1st-graal.md`; `rules-139.md`; `../../client-internals/feature-specs/04-death-respawn.md`
Combat	**1st Graal:** hurt 40 ticks, five knockback steps, sword frames 6–10, reach up to 4 tiles ahead and sideways under 2, sword power a fixed half heart. **1.39 and 2.05:** hurt 40 ticks, knockback 1 tile/tick for 5 ticks (counter 39…35), sword reach 0–4 tiles / sideways < 2. 1.39 hurt pose ~10 ticks then blink; 2.05 uses the `hurt` gani.	`rules-1st-graal.md`; `rules-139.md`; `../../client-internals/decomp-findings-2026-10-24-d/`

What	Value by era	Source
Ledges, push/pull, carry-hide	2.05: jump on the 18th blocked step; `canbepushed`/`canbepulled` from 1.1 beta1, separated in 1.34; hiding under a carried object (mode 14).	`../../client-internals/ledges-push-pull.md`, `../../client-internals/carry-hide-and-glove.md`
Bush, pot and stone respawn	Client-side before 2.00 **[inf]**; server-side from 2.00 ("bush rupees and pot hearts go server-side"). Exact timers are being measured in a separate thread.	**[doc][inf]**
Local multiplayer	Two players from 1st Graal; Two/Three/Four Players to 1.41; removed in NewMain 1.42 / 2.00.	**[bin][live]** start windows
Sleep on start	1.1 starts the offline game with the player asleep in bed (a key wakes the player).	**[live]**
Sound engine	none in 1st–4th Graal; DelphiX DirectSound from 4th alpha (12 named `.wavs`), 29 by 8th Graal, 24–27 named through 2.31; MCI MIDI, plus Winamp control from 1.41 until BASS replaced it; BASS from 2.17 (no Winamp, adds volumes, radio, voice chat); fmod in 3.x–5.x, SDL_mixer on 6.x Linux. Max 8 effects at once (2.05).	**[bin]** survey; `../../client-internals/sound-and-music.md`
Save games	`Graalsav1` … `Graalsv11`; F2/F3 to 1.41; gone in 2.00.	`../../formats/graalsav.md`

Looks: window, screens, HUD and text

What	Value by era	Source
Main window	Caption "Graal". Outer size 648×521 (1st Graal), 648×516 (2nd Graal to 1.1.9), 648×508–509 (1.21–1.39), 648×514 (1.41), 648×507 (NewMain, 2.00), 648×514 (2.01–2.1 tests), 644×507 (2.17–2.31). Form font MS Sans Serif 14 px throughout. 2.05+ add the server name to the title. 2.12–2.16, 2.3 and 3.x have no outer size on record (see below).	**[bin]**
Start window	1st Graal: "Graal - The Adventure", radio buttons One Player / Two Player / One Player Online with host the loopback address and port `14801`. 1.0: "Select the game mode" with the grail picture, character pick per player, "Online At Cyberjoueur" box with port 14802. 1.1 to 1.32: the rider-on-horse picture, mode and character drop-downs; 1.32 adds Language and Nick Name and "Managed by: Stéphane Portha". 1.41: Language, Game mode (One Player … Four Players), Character, Nick Name, "Lead programmer: Stefan Knorr". 2.00: dark-green "Graal 2001" window with the GraalOnline logo, Training / Options / Start. 2.1: new start window and server list. 2.22: Start Online / Training / Options; 2.31: Start Online / Editor / Options. 3.x on: Torque GUI or server-scripted screens.	**[live]** screenshots; **[bin]** forms

What	Value by era	Source
HUD sheet (`state`)	448×96 GIF (1st, 2nd Graal) → 448×137 (3rd–5th) → 298×96 (6th) → 427×109 (7th, 8th) → 427×125/132 (9th, 1.0–1.1.9) → 224×49 (1.21) → 224×81 (1.28b2–1.37, PNG from 1.33) → 224×97 (1.38–1.39) → **224×145 (1.41 to 2.31)**. The NewWorld betas ship the older 224×97 sheet.	**[file]** survey; `../../client-internals/status-display.md`
HUD layout	Spiral (yin-yang) hearts at the top right in every build seen. 1st Graal shows three rupee digits (`000`), 1.0 drops leading zeros (`0`). The A/S/D item buttons appear by 1.32; 1.41 adds the AP bar under them.	**[live]**
Tileset (`pics1`)	1024×512 (1st–4th Graal) → 1280×512 (5th–1.21) → 1856×512 (1.28b2–1.41) → 2048×512 (NewMain 1.42 on).	**[file]**
Name and chat text font	1.3 test adds `font=`/`fontsize=`. 1.38–1.41 executables name Helvetica; the 1.41 install has `font=Helvetica`, `fontsize=24`, the 1.32 install `font=Dialog`, `fontsize=28`. **NewMain 1.42 to 2.31 register the bundled `tempsitc.ttf` (Tempus Sans ITC) at start-up**, and the installs set `font=Tempus Sans ITC`, `fontsize=24`. Live: 1.32 and 1.41 draw NPC names large, bold and white with a dark-blue outline; 2.00 draws them smaller in the rounded face. The chat bubble itself is in a separate thread.	**[bin][file][live]**
Other fonts	Courier New from 1.34 (the F6 debugger), Arial Black in 2.17–2.31 forms.	**[bin]**
Options	Options form from 1st Graal; More Options from 5th Graal; Controls from 15th Graal; player list from 1.28b2; profile from 1.39; server list from 1.41; log window, RPG window and tile-array dialog by 2.17 (no 2.1x binary to place them earlier); 2.31 drops the editor forms (29 → 18).	**[bin]** `builds-deltas.md`
Keys	Arrows or IJKL, A/S/D, M map, Q items, P pause, Tab chat; fixed to 1.37, rebindable from 1.38; F1–F7 and Alt+1–9 per `versions-133-139.md`.	**[bin][doc]**
Heads and characters	Pick list with named characters (Nick, Mandi …) from `heads.gif` in 3rd Graal to 1.1.9; `headsconfig.txt` (Fung Si Yan, Nick, Mandi, Bad Max …) with one image per head from 1.21.	**[file][live]**

Outer window where no form was measured

Date checked: 2026-11-01. Sizes below are only ones already written down. The Decompile folder (client exports and Accessories) has no form-size note and no export of these builds.

Builds	Outer size	Why
2.12, 2.13, 2.14, 2.15, 2.16	not measured	No row in `builds-static-survey.tsv`. `client-internals/drive-clients-inventory.md` records no release between the 2.1 test builds and 2.17.
2.3	not measured	`version-profiles/2.22-2.31.md` says no 2.3 executable was uploaded.

Builds	Outer size	Why
3.04, 3.153	no VCL outer size	The Delphi survey stops at 2.31. These clients are Torque / SDL, not a Delphi form.

The nearest measured Delphi forms sit on either side of 2.12–2.16. 2.1 test 2 and 2.1 test 3 are **648×514**. 2.17.1, 2.22 and 2.31 are **644×507**. **[bin]** `builds-static-survey.tsv`. The change falls between 2.1 test 3 and 2.17.1, so neither size is assigned to 2.12–2.16. 2.22 and 2.31 share 644×507, and that neighbour size is still not assigned to 2.3.

3.04 Linux, in `lab-304-linux-2026-10-30.md`, opens at **800×600** on a 1024×768 display. **[cap]** That sentence records the opening size. It does not split outer size from client area, and it is not a Delphi form size. No window size is written down for 3.153. Torque's minimum screen mode of 640×480 (`torque/engine-and-formats.md`) is a device floor, not an outer window.

What is still unknown

Ranked by how much it changes what a player would notice.

1. **The 1998 to 1.41 rules. 1.39 is done** (`rules-139.md`). **1st Graal is done** (`rules-1st-graal.md`: same five rules, with a fixed half-heart sword, a restart at `house1.graal`, and the death-counter increment left untagged). Still open: 1.32 (Decompile folder re-listed 2026-11-01; still no 1.32 export, so no death, hurt or wall number) and the rest of eras A–B. Walking speed already matches across the measured builds.
2. **2.1 to 2.16 and 2.3.** No binaries. Everything in era F before 2.17 is from release notes and server source. Outer window sizes for 2.12–2.16, 2.3 and 3.x are in the section above: unmeasured, except the 3.04 Linux opening size already in the lab note.
3. **5.x and 6.x look.** The screens come from the server (`Login` server GUI scripts, update packages), so the look is whatever that server sent. `../../protocol/login-session-known.md` records what the existing notes already say that server sends, written with no public capture. The look itself is still unobserved.
4. **Graal3D movement and camera.** Follow distance, look height and field of view are in `../../graal3d/camera.md`. The far plane stays open (library initial back plane 100 **[doc]**; a later DelphiX `SetBack(5000)` was not used; the February 2000 exe was not re-read for `SetBack`).
5. **Default name font before 1.38.** The exes from 1st Graal to 1.37 contain no font family name for the game text; 1.32's install says `Dialog`, which Windows does not have, so a substitute was used.

Files in this folder

File	What
`README.md`	This matrix.
`rules-139.md`	1.39 death, hurt, knockback, sword reach and wall test (matrix unknown #1 for era C).
`rules-1st-graal.md`	1st Graal death, hurt, knockback, sword reach and wall test. 1.32 is recorded as still open.
`builds-static-survey.tsv`	One row per executable: MD5, size, form count, main window, form font, frame timer, sound/config/image name counts, sheet sizes.
`builds-deltas.md`	Per-build additions and removals of config keys, sounds, images, forms and fonts.
`live-runs-2026-10-30.md`	The Wine runs: how, what was measured, screenshots.
`seihai-gaps.md`	Seihai's profiles against this matrix.

The survey script, the form parser, the Wine helpers and the screenshots are in the project folder under `graal-decompile/version-matrix/`, not in this repo.

`version-profiles/matrix/rules-139.md`

1.39 death, hurt, knockback, sword reach and wall test

Date: 2026-10-31. Closes matrix unknown #1 for **1.39** (era C). Tags: **[bin]** from the 1.39 client (plain behaviour only; no addresses), **[lab]** / **[1.39]** from the offline Wine mannerisms run already in `../../client-internals/player-mannerisms-live.md`. Tick = 50 ms.

Compared with the 2.05 rules in `../../client-internals/decomp-findings-2026-10-24/01-death-respawn.md`, `../../client-internals/decomp-findings-2026-10-24-d/01-sword-damage.md` and `../../client-internals/decomp-findings-2026-10-25-k/01-knockback-and-packet-hits.md`.

Summary table

Rule	1.39	2.05 (for contrast)	Tag
Hurt mode length	40 ticks; cannot be hurt again while in mode 3	same	**[bin] [lab]**
Knockback	1 tile per tick while the hurt counter reads 39 down to 35 (five steps); zero vector becomes one tile east; twelve body-box probes; blocked step undone	same five steps	**[bin]** (earlier live table said four tiles; see below)
Hurt pose / blink	pose ~10 ticks then normal; blink starts after the pose, every other tick to 40	`hurt` gani 31 ticks; blink from the hit	**[lab]**
Sword reach	0–4 tiles ahead of the attacker in the facing; sideways offset under 2 tiles	same	**[bin]**
Sword swing length	counter 1→6 (5 ticks); cuts bush	5 ticks (`sword.gani`)	**[lab]**
Death timer	counts 0→30 and stops at 30	same	**[bin]**
Input while dead	any movement/action key respawns once the timer is above 27	same (after 27)	**[bin]**
STATUS dead bit (0x08)	set only when life-state is dead **and** the timer is above 27	same (≥ 28)	**[bin]**
Online default respawn spot	x 30.5, y 30.0 in the current level when no saved entry	same	**[bin]**
Respawn hearts	shared heart-write used on respawn restores to **max hearts**; no AP-band test in this path (1.39 has no AP HUD; bands arrive with 1.41)	AP bands (3 / 5 / max)	**[bin]**
Wall / walk test	six probe offsets per facing step; blocked count pushes at 18 and wraps 38→18; walk 0.5 tile/tick	six probes; same push counters; same speed	**[bin] [lab]**
Baddy touch power	0.5 hearts (1.0 for type 7); within 1 tile on both axes of the baddy centre	same family of powers in 2.05	**[bin]**

Hurt and knockback [bin]

A hit is refused when the player is already in hurt mode (mode 3), in mode 7, or not in the playing life-state. An accepted hit:

1. Clears the current pose sprite.
2. Sets mode 3 and the hurt counter to **40**.
3. Applies the power to hearts (negative power heals; hearts stop at max).
4. Stores a knockback unit vector. A zero vector becomes **one tile east**.
5. If the player was carrying something, may force a drop when the power is positive.

Each playing tick while in mode 3: the counter drops by one; at 0 the mode returns to normal. While the counter is **strictly greater than 34** after that drop (readings 39…35), the player moves one tile along the stored vector, then twelve body-box points (four edges × three) are tested; if any is blocked the step is undone for that tick.

The offline Wine run (`player-mannerisms-live.md`) measured a **40-tick** no-rehit, a ~10-tick hurt pose, blink after the pose, and **four** tiles of knockback on the touch-NPC test. The binary's counter window is the same five-step window as 2.05; the four-tile live figure is kept as a lab observation that may have lost a step to collision or ±1 tick video error. Prefer the binary's five steps for a faithful profile unless a re-run shows otherwise.

Sword reach [bin] [lab]

When another player's swing is judged against the local player, a hit counts only if the victim lies in a facing-relative box: **0 to 4 tiles** ahead of the attacker's centre, and **strictly under 2 tiles** of sideways offset. Power and knockback then go through the shared hurt routine. Offline, S runs a 5-tick sword counter and cuts a bush in front on the first frames (`player-mannerisms-live.md`).

Death and respawn [bin]

When hearts fall to 0 or below:

1. Hearts are clamped to 0; the hurt counter is set to **0**; mode returns to normal; life-state becomes **dead** (value 2).
2. `dead.wav` plays; inventory is partly spilled as floor items (rupees in 100/30/5/1 bands, arrows and bombs similarly).
3. Each tick while dead: if the counter is under **30** it increments by one (so it stops at 30). Between counters **4 and 27** the client cycles death sprites. Above **29** a cleanup helper runs.
4. **STATUS bit 0x08** is included in the prop encoding only when life-state is dead **and** the counter is above **27**.
5. A movement or action key while dead with counter above **27** (or while in life-state 4) runs the respawn routine.

Respawn clears the hurt counter and modes, restores the shared heart field through the same write used when filling hearts to max, warps to the last saved level entry when one exists, and otherwise (online, no saved name) places the player at **x 30.5, y 30.0**. Offline without a saved entry uses the built-in start-level table. No separate "respawn" packet exists on this path.

Wall / walk test [bin] [lab]

Ordinary walking uses a facing step scaled by the move speed (0.5 tile per tick when unburdened; 0.2 when hidden; other multipliers for horse/shield). The candidate position is tested with **six** probe offsets from a facing

table against the tile/NPC block test. A failed probe refuses that step. The blocked-step counter that drives the push pose still uses the **18 / wrap-at-38 → 18** rule already measured live for 1.39. Sliding along a blocking NPC at 0.125 tile/tick was seen live; it is a consequence of the same probe path, not a separate rule.

Knockback reuses the block test with its own twelve-point body box (centre near X+1.5, Y+2.0), matching the later 2.05 layout in spirit.

What Seihai should change

See `seihai-gaps.md` (updated in the same change): for a 1.39 profile, use the 2.05 death timer and sword box, keep hurt lock-out at 40, prefer **five** knockback steps from the binary (not the earlier four-tile live cell), restore hearts to **max** on respawn (no AP bands), and keep the six-probe walk / 18–38 push counters already shared with later builds.

Still open

- Exact numeric contents of the six walk-probe and twelve knockback-probe tables (offsets only; behaviour above does not need them for Seihai's current box tests).
- A fresh Wine knockback distance re-measure to reconcile the five-step binary with the four-tile live cell.
- 1.32 and 1st Graal (next rows of matrix unknown #1).
- Online GServer13 capture of STATUS bit timing and CLAIMPKER (id 14) on a real death; not required to tag the client-side rules above.

`version-profiles/matrix/rules-1st-graal.md`

1st Graal death, hurt, knockback, sword reach and wall test

Date: 2026-11-01. Closes matrix unknown #1 for **1st Graal** (era A). **1.32 stays open** (no decompile; see the end).

Tags: **[bin]** from a static read of the 1st Graal client export (plain behaviour only). The export is the VCL-lossy one already noted in `../../client-internals/early-clients.md`: a few hundred functions, most of them the Delphi library, with the player rules recovered from the routines at the end. Counters are in the 50 ms tick already surveyed for every Delphi build. Where the export's comparisons disagree with themselves, that is called out and not tagged as a number.

Compared with `rules-139.md`. 1st Graal has no AP bar, no ganis, and no later STATUS property.

Summary

Rule	1st Graal	1.39 (for contrast)	Tag
Hurt lock-out	action mode 3 for 40 ticks; a new hit is refused while that mode lasts	same 40	**[bin]**
Knockback steps	one step per tick while the shared counter reads 39 down to 35 (five steps); twelve probes; a blocked step is undone	same five steps	**[bin]**
Knockback direction	baddy and projectile hits store an axis unit step from the attacker's facing; a sword hit stores the centre-to-centre delta	unit vector; a zero vector becomes one tile east	**[bin]**

Rule	1st Graal	1.39 (for contrast)	Tag
Hurt drawing	full hurt body while the counter is above 30; odd counters blink to a single frame	pose ~10 ticks, then blink **[lab]**	**[bin]**
Sword reach	swing frames 6 to 10; ahead by up to 4 tiles; sideways strictly under 2	0–4 ahead, sideways under 2	**[bin]**
Sword power	half a heart, fixed	not a fixed half heart in the later profile	**[bin]**
Death	hearts at or below 0: clamp to 0, clear the counter and the action mode, life-state becomes dead	same clamp; then a 30-tick timer	**[bin]**
Restart after death	a key, once the shared counter is above 27, reloads `house1.graal` at x 24.5, y 30 with hearts and the heart cap both set to 3	saved entry, or (30.5, 30) online; hearts to max	**[bin]**
Wall test	six probes; a blocked step uses the push counter, which wraps from above 37 back to 18; at 18 a ledge test starts an 8-tick hop	six probes; same 18 / wrap-at-38 push counters	**[bin]**

Hurt and knockback [bin]

A hit is refused while the action mode is already 3. Sword hits, baddy touches and the projectile hits below all use one entry:

1. If the player is carrying something, that carry is dropped.
2. Action mode becomes 3 and the shared counter becomes 40.
3. Hearts change by the hit's power (negative hurts). Hearts are a fractional count. They do not rise above the integer heart cap. At or below 0, the death path below runs, and it clears the mode and the counter that this entry just set.
4. A knockback step is stored.

The hurt tick then drops the counter by one. At 0 the action mode returns to normal. While the counter is still above 34 after that drop (readings 39 down to 35), the player moves by the stored step. Twelve points, four groups of three, are tested against the block test; if any is blocked, that tick's step is undone.

Walking is also refused during hurt while the counter is still at least 30. Below that, and until the counter hits 0, walking is allowed again but a new hit is still refused.

Drawing, while the counter is above 30, uses the hurt body. An even counter, or any counter still above 34, draws the full hurt frame list; an odd counter at or below 34 draws a single frame. Once the counter is 30 or below and the mode is still hurt, the normal body is drawn, and odd counters again draw only one frame.

Baddy touch and the common projectile hit store an axis-aligned step of length 1 from the attacker's facing: facing 0 decreases Y, facing 1 decreases X, facing 2 increases Y, facing 3 increases X. A sword hit instead stores the delta between centres (victim minus attacker). The export then scales that delta by its length, but the length adjustment is applied twice to the horizontal component and the vertical component is left at the half-delta, which is the same class of bad assignment this export shows elsewhere. Direction "away from the attacker" is tagged. Unit length for a sword knockback is not.

There is no "zero step becomes one tile east" in these stores.

Powers that reach this entry:

- Sword, and a baddy touch: half a heart.
- A baddy touch counts when both axes are within 1 tile of the body centres (player centre at x+1.5, y+2; baddy centre at x+1, y+1).
- Two projectile types (sprites 65 and 261) hurt by half a heart; the other types in that list hurt by one heart. The same 40-tick mode is used.
- An arrow whose sprite is 107 to 110 is blocked, instead of hurting, when its direction is the opposite of the player's facing and the player's sprite index is below 6. Otherwise an arrow hurts by half a heart.

Sword reach [bin]

Another player's swing is tested only while the local life-state is the normal value. The other player's packed sprite keeps the facing in the low two bits and the frame above that. A hit is possible only on frames 6, 7, 8, 9 and 10.

The box is facing-relative, measured between centres (each centre is x+1.5, y+2). The victim must sit ahead of the attacker, by a span of 4 tiles, and the sideways distance must be strictly under 2. On two facings the export includes both ends of the 4-tile span; on the other two the ends are exclusive. That split is treated as decompiler noise. The tagged reach is "up to 4 tiles ahead, sideways strictly under 2", the same shape as 1.39, and the exact 0 and 4 edges are not tagged.

Power is half a heart, not a sword-power scale. 1st Graal's sword does not grow.

Death and the restart [bin]

When hearts fall to 0 or below, the heart write:

1. Clamps hearts to 0.
2. Clears the shared counter and the action mode.
3. Sets the life-state to dead.
4. Notifies the relay object when one exists. The payload was not read. There is no later-style STATUS property on this client, and this path does not spill the inventory.

The dead life-state changes drawing: the tile buffer is replaced by a filled rectangle, and a text panel is drawn. The body uses one fixed sprite (index 157) only once the shared counter is above 27. Before that, with the action mode already cleared, the normal body is drawn.

A keypress while dead restarts the player once that counter is above 27. The same restart also runs for one other life-state (value 4), with no counter test. The key index handled before this test does not restart; the other indexes do. The restart is the new-game setup, not a saved `Graalsav1` entry:

- Level `house1.graal`.
- Position x 24.5, y 30. The second local player is shifted 12 tiles toward increasing X (x 36.5, same y).
- Facing 2 (increasing Y).
- Hearts set to 3, and the heart cap set to 3, whatever the cap was before.
- Carry, counter and action mode cleared.

The loop that advances the shared counter while dead was not found in this export. Death itself sets the counter to 0, and both the dead sprite and the restart key wait for a reading above 27. A 30-tick cap, of the kind 1.39 has, is therefore not tagged. No death sound is on this path; this build has no sound engine.

Wall and walk test [bin]

The step added for a facing comes from a per-facing table. The length is not an immediate in the routine. The offline run already measured 0.5 tile per tick (`live-runs-2026-10-30.md`); that measurement is not re-done here.

The candidate point is tested with six probes. The block test fails closed: a point is blocked when an object's box contains it, when it lies outside the 64 by 64 board, when either coordinate is below half a tile, or when the tile is not on the passable-tile list. The list itself was not copied out.

If any of the first four probes is blocked, the action mode becomes the push mode and the stepped axis is undone. A fractional part on the free axis is snapped (facings 0 and 2 snap Y down; facings 1 and 3 snap X onto a half-tile). A particular pattern of the six probes instead adds a nudge from a second per-facing table; the nudge offsets were not read. A nudged step clears the push counter. A plain blocked step increments it.

The push counter wraps: a reading above 37 is set back to 18. When the counter is 18 and a ledge-tile test passes, the action mode becomes the hop and the counter becomes 8. The hop then follows a facing table for those 8 ticks and returns to the normal mode. While pushing, once the counter is above 17 and the player is not carrying, the body sprite steps through the push frames.

What Seihai should change

See `seihai-gaps.md`. Seihai has no era A profile. When one is added, the 1.39 / 2.05 combat row is the wrong default: sword power is a fixed half heart, hurt is 40 ticks with five knockback steps, a killing blow goes straight to the dead life-state, and the death key restarts `house1.graal` at (24.5, 30) with 3 hearts rather than restoring a saved cap. The walk test is already the six-probe, push-counter family.

1.32

Not tagged: a listing of the Decompile folder on 2026-11-01, client exports and the Accessories folder, still has no 1.32 export, so no death, hurt or wall number is added.

Still open

- The routine that counts the shared counter up while dead, and whether it stops at 30.
- Exact inclusion of the 0-tile and 4-tile edges of the sword box, and unit length of a sword knockback.
- The six walk-probe offsets, the corner-nudge offsets, and the numeric walk-step table (the 0.5 tile step is the live measurement only).
- 1.32 for all five rules, and the rest of eras A and B.

`version-profiles/matrix/seihai-gaps.md`

Seihai's version profiles against the matrix

Date: 2026-10-30. Compared: Seihai `main` at 72ed21d (`client/src/protocol/profile.rs`, `chain.rs`, `documentation/version-profiles.md`) with `README.md` in this folder.

What Seihai already covers

Seihai has 26 profile rows, 1.33 to 2.31, built as a version chain (1.33 written out, each later build a delta). A row sets the token, client type, cipher generation, listserver endpoint and login shape, the server→client and client→server id sets, prop limits, file and level features, the death and combat rules where traced, and UI switches (server list style, chat recall, HUD sheet layout, minimap, editor, ganis, emoticons, key config, log window, ghost mode, server name in the title, GS1 count). The 50 ms game clock, the half-tile walking step and the 8-voice sound cap are global and match every build measured (`live-runs-2026-10-30.md`), so they need no per-version switch.

Missing, by size

1. Eras with no profile at all

Era	Builds	What a profile would need
Z, A: 1998 (Java and 1st Graal to 1.1.9)	21	The gen-1 relay (lobby `N/P/C/D` lines, game names `Zelda2`, `Graal` … `Graal8`, `D` payload types per build), `.zelda` and GR-V1.00–1.03 levels, `Graalsav1`–`8`, GIF sheets of their own sizes, the 1998 start windows, two-player local mode. `protocol/relay-1998.md`, `formats/graalsav.md`. `EncryptionGen::Gen1` exists in Seihai but no row uses it.
B: 1.21 to 1.32	5	GSERV007–018 login on gen 2 (`protocol/gserv-1999.md`), port 14804 then 14900, the 1.2x id ranges, `Graalsv11` saves, 224×49 and 224×81 HUD sheets.
3D0: Graal3D (2000)	1	Its own client and server protocol (`graal3d/protocol.md`); a separate program, so it fits the planned Seihai suite rather than the 2D selector.
H: 3.x	2	Gen 5 as 2.22/2.31, but login tokens `G3D03014`/`G3D28015`, no tamper check, the signed `license.graal` list host, the category prefix that picks a server-list tab, 79/88 handled ids. Close to 2.22/2.31 on the wire.
I, J, K: 4.x to 6.x	about 20	Login id 5 with the machine-id list, the `Login` game server instead of a listserver, update packages, GS2 bytecode and server-built GUI; 6.11's script-defined framing. A different client rather than a profile row.

2. Look switches the rows do not have

Gap	Matrix value	Seihai today
Name and chat text font	Bundled Tempus Sans ITC at 24 from NewMain 1.42 to 2.31; Helvetica-named before (1.38–1.41); `font=`/`fontsize=` from 1.3 test	One 5×7 bitmap font for every version (`render/font.rs`). The chat-bubble thread covers the bubble itself.
Start window	Distinct per era (see the matrix and the screenshots)	One launcher for every version
Main window size	648×508 (1.3x) / 648×514 (1.41, 2.01–2.1) / 648×507 (2.00) / 644×507 (2.17+). 2.12–2.16 and 2.3 were not measured. 3.x has no VCL outer size; 3.04 Linux opens at 800×600 in the lab note, with no outer-versus-client split	Not per version

Gap	Matrix value	Seihai today
HUD sheet layout	224×81 (1.33–1.37), 224×97 (1.38–1.39), **224×145 (1.41 to 2.31)**	`HudArt::BlankButtons` covers 1.38 to 2.31 and its comment calls the later builds inferred. The HUD already draws the AP and MP bars only when the sheet has their frames, so the drawing is right; the comment is not. (Fixed in the accompanying Seihai change.)
Options and windows	Per-build form lists (`builds-deltas.md`)	Not modelled
Local multiplayer and save games	Up to 1.41	Not modelled

3. Rules that are only partly per version

- **Death and combat.** 1.41 carries the 2.05 rules from the template, 1.42 to 2.04 carry none, 2.05 and later carry them (`chain.rs` already lists this as unknown). **1.39 is now traced** (`rules-139.md`): same 30-tick death timer, STATUS bit after counter > 27, 40-tick hurt lock-out, five knockback steps, sword box 0–4 / <2, six-probe walk. Differences for a 1.39 row: restore hearts to **max** on respawn (no AP bands; the AP HUD starts in 1.41), and the hurt pose is a short built-in animation (~10 ticks) rather than the later `hurt` gani. **1st Graal is traced** (`rules-1st-graal.md`) and must not copy that row: sword power is a fixed half heart on frames 6–10, hurt is 40 ticks with five knockback steps, and the death key restarts `house1.graal` at (24.5, 30) with hearts and the heart cap both set to 3. Seihai still has no era A profile to hang this on. 1.32 is still open (no decompile).
- **Bush, pot and stone respawn.** Client-side before 2.00, server-side after (release notes). The exact timers are being measured in another thread.
- **Tamper check.** Seihai has the per-build constants and resend rules opt-in; nothing new here.

4. Assets per version

Seihai reads each version's own asset folder (`assets.<label>=`). The matrix lists which sheets and sizes each build shipped (`builds-static-survey.tsv`). The archives for 2.00 to 2.05 hold only the exe, so those versions need the art from a neighbouring install; 1.41 rev1 and NewMain 1.42 (`graalnm26`) are the closest full installs.

Suggested order

1. Name font per version (Tempus Sans ITC for 1.42+), since every screen shows it.
2. A 1.21–1.32 profile (GSERV login on gen 2 is a small step from Seihai's gen-2 1.33 path).
3. The 1998 relay as its own mode.
4. 3.x as two rows next to 2.22/2.31.
5. 4.x and later as a separate client in the planned suite.

`version-profiles/version-coverage-2026-10-24.md`

Version coverage: is Graal_analysis enough for Seihai's multi-version selector?

Date: 2026-10-24 (Asia/Tokyo). Read-only review. No pushes, no PRs, no public connections.

> **Update (same day, after loopback lab runs with real 2.17, 2.22 and 2.31 clients):** the tampercheck gap below is closed. The constants for all three builds are in `tampercheck-values.tsv`, and the payload is 10 raw bytes, not two 32-bit words. The 2.17 listserver flow, the 2.22 gmap edge crossing, the mute payload and the client's short compressed gen-5 frames are now documented in `protocol/listserver.md`, `client-internals/feature-specs/06-gmap-overworld.md`, `.../14-`

> `friends-ignore.md` and `protocol/codec-and-framing.md`. 2.17 and 2.31 were accepted by the lab gserver with the merged Seihai. The table and gap list below are left as the original snapshot. Reviewed: `squirelworks/Graal_analysis` main at 8da6800 (PRs #1 to #26 merged), `squirelworks/seihai` main at 3383300, plus the lab handoff notes and lab run directory (including the 2026-10-24 public 2.22 report) and `seihai_notes/`. Only derived facts here: no decompile text, addresses, or internal function/data names.

Short answer

Yes for the protocol skeleton, no for full per-version behaviour. Graal_analysis (PRs #24, #25, #26: `client-internals/version-profiles/*.toml` + `resolve.py`, `version-profiles/2.22-2.31.*`, `protocol/client-versions.md`, `version-tokens.md`, `version-id-map.md`) now has, per version, a token, client type, generation, listserver login, server-to-client id set, client-to-server id set, prop ranges, file-transfer flavour and level formats, each tagged bin / cap / srv / doc / inf. That is enough to *select* 1.33 through 2.31 and log in with the right token, cipher and framing. Seihai already has the abstraction (`ClientVersion`, 23 rows in `profile.rs`, `EncryptionGen` gates login and framing, `plo`/`pli` id sets gate dispatch and sends).

What is missing is mostly **payload layouts of the newer packets, the tampercheck content, and live confirmation** of everything except 2.05 and 2.22.

What Graal_analysis covers

Topic	Where	Coverage
Tokens, build to token	`version-tokens.md`, `version-id-map.md`	Complete 1.21 to 2.31 plus 3.x. 2.10-2.16 from gs2lib only
Game-server login (type, key byte, token, account, password)	`client-versions.md`, `login-and-walk.md`	2.05, 2.17, 2.22, 2.31 from binaries. Older rows inherit
Generations	`codec-and-framing.md`	gen 2, 3 (2.05 verified on 704 messages), 5 (2.22 verified). Gen 4 (2.19-2.21) server-only
Gen-5 client send rules	`client-versions.md`	Size thresholds, XOR limits, queue cap
Listserver	`listserver.md`	2.05 (id 0), 2.17 (id 0 then id 1, host from signed `arrays2.dat`), 2.22/2.31 (id 7 + `GNW30123` + gen 5)
PLO set per version	profile TOMLs	Read from binaries for 1.33-2.05 (24 exes), 2.17, 2.22, 2.31. Inferred for 2.1-2.16, 2.3
PLI set per version	profile TOMLs	Binary for 1.41-2.05, 2.17, 2.22, 2.31; 2.22 has 3 unmatched ids (36, 42, 45). Inferred 2.1-2.16
Packet id tables with payloads	`packet-ids.md`, `packet-layouts.md`	Names for all ids. Payload layouts good for 0-48 and 2.05 senders. Thin for 49 and 168-189
Props (player, NPC, baddy)	`player-props.md`, TOML `props_max`	Good to 2.31 (X2/Y2/Z2 encoding documented). Other-player parser for 2.17 not read
File transfer	`file-transfer.md`	Complete for pre-2.1, 2.1-2.13, 2.14+. Large-file chunking from server source only
Levels	`level-binary*.md`, `glevnw01.txt`, `gmap.md`, `v3x-clients.md`	GR-V1.00-1.04, GLEVNW01, GRMAP001, `.gmap`. Per-version loader lists in TOML
Version differences (player-visible)	`version-profiles/README.md`, `2.22-2.31.md`, `feature-timeline.md`	Rich. HUD, windows, keys, GS1 command counts
UDP	`udp-p2p.md`	2.05 captured. Other versions inherit

Seihai side: how versions are selected and gated

- `protocol/version.rs`: `ClientVersion` enum (1.33-1.39, 1.41, 1.42, 2.00, 2.01, 2.02, 2.05, 2.1, 2.12-2.16, 2.17, 2.22, 2.3, 2.31). `EncryptionGen` = Gen1/2/3/5.
- `protocol/profile.rs`: one `VersionProfile` row per version, built from templates `CLASSIC`, `BASE` (2.05), `NEW_2_1`, `GEN5`. Fields: client type, generation, listserver port and token, `plo`/`pli` IdSets, language and tampercheck flags, prop sets, file flags, level formats, UI flags, `Basis::{Binary,Inferred}`.
- `net/mod.rs`: `login_versioned` and `login_listserver` branch on `EncryptionGen`; sends are refused when the id is not in the row's `pli`; dispatch drops ids not in `plo`.
- The rows were written from the **earlier** notes (cross-version-timeline, client-versions), before the merged TOML profiles. So Seihai and the TOML disagree in places (below).

Per-version table

(TSV copy: `version-coverage-2026-10-24.tsv`.)

Version	Handshake known	Encryption / framing known	Packet table known	Gaps	Verdict
2.00 (`GNW24011`)	Yes, bin: type 0, key byte, token, gen 3. List host `<list-host-classic>:14921` DES-hidden (dead; redirect needed)	Yes, gen 3 as 2.05	PLO 51+2 ids and PLI from binary (4 revs, no differences). Prop parse 0-36	Token absent from gs2lib: stock GServer-v2 treats it as unknown, so lab needs an allow-list patch. Login props unverified. Never run live. Seihai row reuses the 1.41 PLI set (includes 35/36 which first appear in 2.01)	**Partly**: enough to implement, untested
2.05 rev 2 (`GNW03081`)	Yes, bin + lab	Yes, gen 3, 704 messages	Yes: PLO 59, PLI 0-24/26-30/32-38/80/81	Port and host per listserver (built-in 14921 vs Reborn 14922/14923); lab patch maps it onto 2.1 behaviour; tampercheck absent	**Enough**
2.01-2.04 (`GNW23021`, `GNW15041`, `GNW20041`, `GNW06051`, `GNW24061`)	Token yes (catalogue); login as 2.05	Gen 3	Binary for PLO/PLI deltas per build	Seihai has no 2.02 rev2, 2.03, 2.04 rows (language at 2.02 rev2, trigger at 2.03 not modelled); live untested	**Partly**
2.1-2.16 (`GNW31101` ... `GNW12102`)	Tokens yes (gs2lib). Login inferred as gen 3	Gen 3 inferred (2.17 is gen 3)	**Inferred** from neighbours and release notes	No binary of any 2.1x release in the archive. 2.1 test builds still send 2.05's token	**Needs binary or capture**; low priority, can ride on 2.05 or 2.17

Version	Handshake known	Encryption / framing known	Packet table known	Gaps	Verdict
2.17 (`GNW22122`, 2.171)	Yes, bin: type 0, gen 3, list via `arrays2.dat` (id 0 then id 1, host and port from file; port default 14922)	Yes, gen 3	PLO 79, PLI 61 ids from binary; PLO 169-185 present	TAMPERCHECK (39) content untraced (two 32-bit words); other-player prop parser and GS1 count not read; not run live; Seihai `pli` for 2.17 omits 40, 41, 43, 44, 45 and its list port default is 14921	**Partly** (login/ framing enough; behaviour thin)
2.22 (`GNW03014`; list `GNW30123`)	Yes, bin + lab + 4 public gservers + public listserver	Yes, bin + lab + public: 27/27 gserver frames and both list frames decode	PLO 80, PLI from binary (36, 42, 45 unmatched)	Payloads for ids 25, 39, 41, 44, 47, 156, 168, 174, 179, 180, 182 (sent by every public server) are names only; tampercheck is a 10-byte per-build constant (lab value known, not sent by Seihai; public servers did not kick within 20 s); client bz2 send path never seen; Seihai `pli` for 2.22 = 2.17 set (allows 36/38/81, refuses 40/41/43/44)	**Enough** to log in and play; **partly** for full parity
2.3 (`GNW14015`)	Token yes; everything else inferred from 2.31	Inferred (gen 5)	Inferred	No binary exists	**Partly** (treat as 2.31)
2.31 (`GNW28015`)	Yes, bin: same code as 2.22, `GNW30123` for list	Yes, bin (same thresholds, XOR limits, queue cap)	PLO 81, PLI = 2.22 minus 21 plus 46	Never run (lab or public); stock GServer-v2 has the token commented out; `MOVE2` (189), `LARGEFILESIZE` (84), `ENTERLEVEL` (46) layouts not documented; tampercheck value unknown; X2/Y2/Z2 encoding documented	**Partly**: one lab capture away from enough
1.33-1.39 / 1.41 / 1.42	Yes (bin, 1.39 lab-verified as gen 2)	Gen 2 (1.33-1.39), gen 3 (1.41+)	Binary per build	1.41 list host unknown; no listserver before 1.41	**Enough** for a selectable row

Concrete gaps, ranked

1. **Payload layouts of ids 49 and 168-189 (and 25, 39, 41, 44, 47, 156)** are names or one-liners in `packet-ids.md`. Needed for bigmap/minimap, ghost mode, serverwarp, RPG window, status list, process list, level heights, BOARDMODIFY2, MOVE2, LARGEFILESIZE. *Fillable now:* private public-server captures

already on the lab box (never committed; all frames decoded) show most of these ids; the 2.17/2.22/2.31 exes + `.idc` name lists cover the handler arms for a cleaned note. 2.00 export is not needed.

2. **PLI_TAMPERCHECK (39) content** for 2.17, 2.22, 2.31. 2.22 has a lab constant in the TOML; 2.17 and 2.31 are unknown. *Fillable from* a static read of the builder in the `.exe`/`.idc` (2.171, 2.222, 2.31), or a lab capture of each real client under Wine. Needed before a stricter server enforces it.
3. **Seihai rows lag the merged TOMLs.** 2.22/2.31 `pli` should follow the binary list (2.22: no 36/38/81, adds 40/41/43/44; 2.31: minus 21 plus 46); 2.17 `pli` should add 34-36, 40, 41, 43-45; 2.00 should not send 35/36; no rows for 2.02 rev2, 2.03, 2.04; 2.17 listserver port. *Fillable now*: no decompile needed. Wire `resolve.py` output into `profile.rs` (or a generated table).
4. **Doc conflict on the gen-5 XOR group rule.** `codec-and-framing.md` and `client-versions.md` still say `floor(n/4)` capped at 4 for types 4 and 6. Lab (610 frames) and public (27 frames) show one step per group present, a partial last group included. The doc needs correcting, with a note of which side (client send vs server) was read. Also the client-originated bz2 frame (>= 0x1000) is unseen. *Fillable now* from existing captures and the 2.22 `.idc`.
5. **2.1-2.16 and 2.3 have no binary.** Rows are inferred. *Needs new binaries*; nothing on the box covers them. Practical fix: mark them as aliases of 2.05 (2.1 line) or 2.31 (2.3) in the UI.
6. **Live acceptance is unknown for everything except 2.22.** Public Reborn servers may reject 2.17, 2.31, 2.05 or 2.00 tokens, and may treat a missing tampercheck differently. Stock GServer-v2 allows 2.22 only by default. *Needs captures*: lab runs for 2.00 (allow-list patch), 2.17, 2.31 with the real Wine clients; public runs need explicit approval.
7. **Listserver per version.** 2.17 id 0 + id 1 against the Reborn listserver on 14922 is untested; 2.05/2.00 depend on a host redirect. Port should be per-profile and overridable. Needs one capture, not a decompile.
8. **Smaller:** 2.22 PLI ids 36, 42, 45 call sites unmatched; 2.17 other-player prop parser unread; 2.22 `#R` GS1 code meaning; 2.1-2.16 GS1 counts.

Which gaps the existing exports can fill

Source on the box	Fills
`donors/Graal_2171.exe` + `.idc` + `.names.tsv`	2.17 tampercheck builder, other-player prop parser, PLI send sites, `arrays2.dat` handling
`donors/Graal_222.exe` + `.idc`	Payload layouts for ids above 48, PLI sites 36/42/45, tampercheck, client bz2 path
`donors/Graal_231.exe` + `.idc`	`MOVE2`, `LARGEFILESIZE`, `ENTERLEVEL`, 2.31 tampercheck
`donors/Graal_v2_00.c` (2.00 export, 193k lines)	2.00 PLI set cross-check, login props, 2.00 level and prop details
Existing public pcaps (box only, never commit)	Payload shapes of 25, 39, 41, 44, 47, 156, 168, 174, 179, 180, 182 and the bigmap/minimap messages
New captures	Anything about 2.31 and 2.17 live behaviour, 2.00 against a patched lab server, other-player parser behaviour, public acceptance of tokens
New binaries	2.1 to 2.16, 2.3

Note: the `.idc` files are name tables. Reading the handler bodies still needs a disassembler pass (Ghidra) over the matching `.exe`.

Recommendation

1. Treat 2.05, 2.22 as the verified baseline; 2.17 and 2.31 as selectable now with "partly" labels.
2. Regenerate the Seihai rows from the TOML (gap 3), then fix the XOR note (gap 4). Both are doc/data work with no new decompile.
3. Run one Ghidra pass each over 2.22 and 2.31 for gaps 1 and 2, write a cleaned note per version, then lab-capture real 2.17 and 2.31 clients to confirm.

4. Ask for approval before any public run for 2.17, 2.31 or 2.05.

Status update, 2026-10-24 (decompile pass of 2.00, 2.17.1, 2.22, 2.31 and the Drive clients set)

- **Gap 1 (payload layouts of ids above 48 and 25, 39, 41, 44, 47, 156):** closed for 2.17.1, 2.22 and 2.31, except one small item (the extra step 2.31 adds in 156; the `PLO_SERVERWARP` field meanings and the `PLO_BOARDMODIFY2` trailer were settled by lab runs). See `protocol/packet-layouts.md` and `plo-layouts-217-231.tsv`.
- **Gap 2 (tamper-check content):** the lab values for 2.17, 2.22 and 2.31 are in `tampercheck-values.tsv`, and the builder is now understood: two chained CRC-32 words over the client's own memory image (`tampercheck-words.tsv`; `client-internals/decomp-findings-2026-10-24-b/01-tamper-check-id39.md`). Whether any server enforces it is still unknown.
- **Gap 6 (2.00):** a real 2.00 exe and three revisions are in the Drive clients set. The "allow-list patch" wording above means a config entry only: the lab server needs `GNW24011` in its allowed-versions list, no code patch (`client-internals/decomp-findings-2026-10-24-b/04-v2.00-client.md`). The 2.00 client still needs its list host redirected to the lab listserver, and its listserver frames have not been exercised.
- **Gap 5 (2.1 to 2.16 and 2.3):** unchanged. The Drive clients set has no 2.1.x to 2.3 builds, only the two 2.1 test builds (`client-internals/drive-clients-inventory.md`).
- **Gap 8:** client to server ids 36, 42 and 45 of 2.22 are still unmatched to call sites.
- Per-build handled server-to-client ids for 2.00 to 2.31 are now tabulated in `plo-handled-ids-by-build.tsv`, and the client-to-server ids per 2.0x build in `pli-sent-ids-by-build-20x.tsv` (2.00 lacks 35 to 38).
- **Update (lab batch b2):** the 2.00 listserver step is partly observed (client request seen; the list reply it accepted was synthetic) and the id 46 and id 186 trailer questions are settled. See `client-internals/decomp-findings-2026-10-24-b/07-lab-findings-46-186-200.md`.
- **Update (lab batch c):** `PLI_ENTERLEVEL` (46) is confirmed on doors, links and server warps in 2.31 and absent in 2.22; `PLI_SHOWIMGPLAYER` (24) is relayed unchanged and 2.22 sends it too with two players; `PLO_SERVERWARP` (178) field order is settled for 2.31. See `client-internals/decomp-findings-2026-10-24-b/08-lab-findings-levelchange-24-178.md`.